AGAPPE
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USER MANUAL



MISPA HX50

Automatic Hematology Analyzer

SYSTEM VERSION COPYRIGHT

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4. The protection supplied by the instrument might be broken if users do not use the instrument according to the regulated methods in the manual.

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Introduction

Thanks for purchasing Automatic Hematology Analyzer (hereinafter referred to as "Analyzer") of Agappe Diagnostics (hereinafter referred to as "The company")

■ Please use the analyzer under the supervision of technicians who have received professional training. Please use it under the supervision of physicians or clinical laboratory technicians when it is used for clinical examination.
■ In case of incorrect use of reagents or consumables, may generate wrong result, and even cause damage to analyzer or operator.
■ Please read the user manual through before use for correct operation of the instrument.
■ Please keep the manual to a place with an easy access for future reference.

Contents of manual

This manual contain 12 chapters and 1 annex. Please search for exact chapter according to requirement.

Chapter	Chapter content
Chapter 1 Overview of manual	■ Summarize the analyzer manual
Chapter 2 System Introduction	■ Introduce the intended use and measurement parameter of analyzer ■ Introduce the constitute and operation interface of analyzer
Chapter 3 Design principle	■ Introduce the design principle and working process of analyzer
Chapter 4 System Installation	■ Introduce the installation requirements and method of Analyzer
Chapter 5 Basic Setting	■ Introduce the reference range and other system setting of analyzer
Chapter 6 Routine operation	■ Introduce the sample collection, preparation and analysis process ■ Complete the routine operation of analyzer
Chapter 7 Work	■ Sample input, results view and any other operation in Work interface
Chapter 8 Results View	■ View the sample analysis results of analyzer
Chapter 9 Analyzer Calibration	■ Introduce the calibration requirements and method of analyzer
Chapter 10 Quality Control	■ Introduce the quality requirements and method of analyzer
Chapter 11 Maintenance	■ Introduce the maintenance method of analyzer
Chapter 12 Troubleshooting	■ Introduce the fault cause and handling method of analyzer.
Annex	■ Introduce the specification of analyzer

Product guarantee

Due to limited content in the user manual, please refer to the methods recorded in the user manual to ensure normal use of Automatic Hematology Analyzer.

a. Range of guarantee

The instrument will be repaired free of charge if it malfunctions due to a defect in manufacture. But in the case of repairs, sometimes substitute or replacement of the item will be exchanged.

b. Guarantee territory Within the country of sales.

c. Disclaimers

Guarantee is not applicable even in the guarantee period in case of the following:

- Failure due to use under improper installation condition.
- Failure caused by the power supply, voltage or frequency that is not stated in the manual, or caused by abnormal power supply.
- Failure caused by circuit corrosion and degradation that because of strong corrosive gas in air.
- Failures caused by use of hardware, software or accessories not provided by the company.
- Failures caused by wrong application method or application method not recorded in the manual, or failure due to the maintenance not conducted by Agappe.
- Failures due to maintenance by companies not designated by Agappe.
- Failure due to movement or transportation which not approved by Agappe.
- Failure resulting from dismantling or modification by customer.
- Failure caused by fire, earthquake, wind, flood, thunderstrike, crime,
- violence, terrorist, radioactive contamination, warn, hazardous substance
- and any other irresistible natural hazards.
- Failures occurring in obsolete instrument or instrument repurchased without written permission from Agappe.

d. Rights on the use manual

This user manual may be modified without advance notice.

Installation, move, and after-sale service

a. Installation and move

During installation, user needs to prepare hardware conditions for installation of the instrument and appropriate working environment based on the user manual.

b. After-sale service

- Please contact the service department of company or authorized distributor regarding after-sale service and maintenance contract.
- Please bring the maintenance contract, inspection service contract and so forth to negotiate with the company or the service department of authorized distributor on the service beyond guarantee period.

Disposal

Disposal of instrument need obeying relevant national regulation.

How to get help

If you encounter any problems and need help during installation and use of Automatic Hematology Analyzer, please contact the service department of AGAPPE. or authorized distributor with the contact methods as below:

- Company website: www.agappe.com

Please prepare the following information:

- Production number of Automatic Hematology Analyzer (check the nameplate on the backboard of the instrument)
- Description of problems
- Methods and operation steps you have used to solve the problem
- Your contact information, including your mobile number, fax, and email address, etc.

User training

The company will provide training service on operation of Automatic Hematology Analyzer. Please consult the service department of the company for detailed information.

Production date and shelf life

The production date of instrument is labeled on nameplate, the shelf life is 5 years.

Disposal of the instrument and liquid waste

Before disposing the instrument, please contact our service department for advice on safety procedures.
Please dispose the instrument according to local environment protection regulations.
Please dispose the waste according to applicable environment protection regulations strictly.

Storage and transportation

Storage

Instrument with package should be stored in the condition of no corrosive gas and well-ventilated environment, and temperature is -10*C ~ 40。, relative humidity is 10% ~ 90%.

Transportation

Please use the designated special packing box of the company for transportation, and prohibit transportation without packing.
The packing box should be fastened to the conveyance according to packing instruction, prohibit dumping and impacting.
Pay attention to moisture and sunscreen and prohibit rain during transportation.

Other precautions

(1)Environmental conditions

Instrument should be used when meet the following environmental conditions.

a. Clean air and well-ventilate.
b. Avoid direct sunlight.
c. The table keeps horizontal (slope is 1/200 or lower)
d. The table can withstand the weight of instrument.
e. Ambient temperature is within the range of 15°C ∽30°C , when measuring, the error is ±2 .
f. Relative humidity is within the range of 30% ~ 85%, and there is no condensation on the surface of the instrument
g. No vibrate that can be felt by human body.
h. There is a power distribution board within 5 meters of instrument.
i. Voltage fluctuation is within the range of ±10%.
j. Please keep away from the devices which can produce the electromagnetic and magnetic interference (eg. centrifuge, EDM machine, telephone, transceiver and cordless phone etc).
k. There is a ground terminal (grounding resistance is 10Q or lower).
l. Noise is less than 60db when instrument is running (not include occasional noise).
m. Noise is less than 45db when the instrument is in readystatus.

(2) Disposal of reagent, sample and consumables

a. Please follow the measures described in national safety guidelines or regulations for biological hazards, or relevant environment protection regulations in the country of sales.
b. Users are responsible for disposal, preservation or abandonment after use of the samples according to given standards. Therefore, it is recommended to take out and store a part of samples beforehand to avoid unexpected loss of the sample due to operation mistake or other reasons.

Important safety information

Please read carefully the following warnings and reminders before operation of Automatic Hematology Analyzer.

The warning signs on the instrument or the danger warnings in the manual contain the following warning titles composed of warning signs and single words (like danger, warning and caution).



	A safety warning symbol. It appears before the word in a danger warning in relevant safety descriptions of this manual. Please pay attention to all the safety information behind this symbol.
	Danger: It indicates emergent dangerous situation, which would cause several injuries or hurts if it cannot be avoided.
	Warning: It indicates potential dangerous situation which would cause several injuries or hurts if it cannot be avoided.



Caution: It indicates potential dangerous situation which would
cause mild or moderate hurts if it cannot be avoided.



Caution: It indicates potential dangerous situation which would cause loss of property and data, or cause environment pollution if it cannot be avoided.



Danger is not used in this manual.

In addition to the warnings and reminders mentioned above, the following words will be used for expression in this manual.

Note: It indicates the directions for use which to ensure proper use of the product for correct test so as to prevent from any failure.

Precautions before operation

Instrument operation and user manual

Please make sure to put on protective clothes before use of the instrument so as to avoid contamination from infectious substances. Protective clothes are important more than ever in maintenance and inspection. Please obey the regulations if there is specific regulation or handbook.
Do not carry out any operation except mentioned in the user manual.
Please contact the service department of the company if there are any problems with the instrument.
Please be sure to comply with the precautions and directions described on the instrument or in the user manual. The instrument might be broken if users do not use the instrument according to the regulations in the manual.
Not all the accidents are predicable, so the dangerous warning signs in the manual and on the instrument cannot cover all the possible situations. It is not enough to just follow the given directions. Please be on the alert and use the common sense.

Instrument Operation

Please test the control during inspection of the reagent stability and instrument operation to ensure control of the accuracy. Before testing of patient sample, please make sure the test result of the control meets the requirements. Please obey the requirements of user manual about the storage, disposal and use method of some reagent and control (before and after unpacking).

Other safety preventive measures

Please do not modify the instrument or use components except those permitted in the manual, otherwise, danger may occur.
Please avoid to use other organic reagents other than used for maintenance.

Precautions for use

In the use of this instrument, please pay attention to abnormal noise, water leakage, and other abnormalities. If the above problems occurred, please take proper safety measures according to specific conditions and contact the service department of the company.
Please don't open the top cover plate of the instrument during operation, otherwise, danger may occur.

Precautions on installation and maintenance

Users are not allowed to install the instrument individually. To ensure the safety and normal use of the instrument, the service department of the company will provide installation service.
Assays under user's maintenance are restricted to what is described in this manual. Please comprehend the details before maintenance. Please don't try to maintain assays not described in the manual; otherwise, the instrument will be put in danger and personal injury may occur.
If the warning signs are obsolete or damaged, please contact the service department of the company to replace it.

Warning notices

The warning notices in this manual and the positions are listed below. The protective devices mentioned below includes medical gloves, protective glasses, protective cloths, and mask.

Warning notices List



Infection caused by liquid waste

- Contact with the liquid waste may cause infection.
- Please put on the protective devices before maintenance and inspection.
- If the liquid waste contacts the skin, please wash the skin immediately with flowing water and disinfect the contact site. If necessary, please seek medical advice.



Infection caused by the sample

- Contact with the sample may cause infection.

- Please put on the protective devices when it disposing the sample. If the
- sample contacts the skin, please wash immediately the skin with flowing water and disinfect the contact site. If necessary, please ask for medical advice.
- If the sample drops onto an instrument, please put on the protective device, wipe off the solution and disinfect the contact site.

Infection/ damage caused after contact with instrument units

- Contact with sampling unit, reagent injecting unit, cleaning unit or other units may cause infection or damage.
- Please make sure to put on the protective device and abide by the procedures described in the manual at the time of maintenance and inspection.
- Please make sure that the instrument is not in operation before maintenance.
- Contact with the sample rack and sample tray may cause injury or infection.
- Please load or unload the sample at the indication of sampling stop by the software when the instrument is in operation. Before loading or unloading the sample, please rotate the sample tray to the safe position.
- If there is an alarm on the sample arm and the sample tray, please do not load or unload the sample even if the sampling stops.

- **Skin irritation caused by acid or alkaline chemicals**
- Contact with substrate or wash buffer may cause skin irritation.
- Please make sure to put on the protective device and follow the precautions indicated in the reagent instruction at the time of disposing the substrate or wash buffer.
- If the substrate or wash buffer contacts the skin, please wash the skin immediately with flowing water and disinfect the contact site. If necessary, please ask for medical advice
- Please make sure to put on the protective device and follow the precautions indicated in the reagent instruction during disposing the reagent.

Electric shock caused by high voltage

- There is a risk of electric shock because of some parts of the detecting unit are under high voltage.
- Do not remove the cover plate from the detecting module.

Burn injury caused by fire

- Dyeing reagent is a flammable substance, if not handled properly, it is easy
  to cause fire or bums.
- Do not get close to the fire during staining the cells by dyeing reagent.
- The dyeing reagent is used for cell staining, and user is responsible for

Notice List

Infection caused by improper disposal of liquid waste

- Liquid waste which contains infectious pathogens (blood samples) should be disposed according to relative laws and regulations by manager who takes charge of the management of infectious medical wastes.
- The substance which contained in reagent, calibrators and control are restricted by environment protection laws, and disposal of them should obey the local quality standards on waste water.

Improper disposal of infectious waste

- Waste materials belongs to infectious wastes, which should be disposed according to relative laws and safety control regulations by manager who takes charge of the management of infectious medical wastes.

Inaccurate measurement caused by insufficient sample drain

- If samples contain insoluble substances, such as fibrous protein or dust, the test results will be inaccurate because of the insufficient sample absorption. Please make sure that there are not insoluble substances in the samples before loading the samples.
- If air bubbles appearing in the sample container, the test results will be inaccurate because of the insufficient sample absorption. Please make sure that there are no air bubbles in the samples before loading the samples.

Inaccurate measurement caused by improper reagent injection

- If air bubbles appeared in reagent bottle, the test results will be inaccurate because of the insufficient sample absorption.
- Please make sure the reagent do not have bubbles before loading reagent.

Hurt caused by working compartment cover extrusion

- During open/close procedure, caution operation should be performed to prevent from hurt

Reagent absorption by mistake caused by manual registration

- If the barcode of the reagent is wrongly input during manual registration, the test result will be inaccurate as a result of using wrong reagent.
- During reagent registration with manual system, please correctly input the bar code on the reagent bar code label based on the procedure indicated in this user manual.

Misreading of the calibrators or control caused by manual registration

- If the bar code of the calibrators or control is wrongly input during manual registration, the test results will be inaccurate because of using the wrong

calibrators or control.

- Please correctly input the bar code on the bar code label of the calibrator or control based on the procedure indicated in this user manual during reagent registration with manual system.

Fatigue caused by long-time operation

- You may have increasing fatigue in body and eok if you are working before LCD with the same gesture for a long time. For your health purpose, please take a break of 10-15 minutes after every one hour of continuously working before LCD. It cannot exceed 6 hours daily for a long time staring at the LCD.

Warning signs

The following are warning signs labeled on Automatic Hematology Analyzer. Please read each warning signs carefully and confirm the contents on corresponding components. Please check the warning signs regularly, and implement cleaning and maintenance to make sure they can be clearly read within safe distances. If the warning signs are illegible because of ageing, please contact the service department of the company for replacement.

Signs description

Warning It indicates that a hazard which will cause severe injury or death in the adjacent area of this sign. Please make sure to read this user manual carefully for proper use.	
Biohazard Warning It indicates that a biohazard in the adjacent area of this sign. Please be sure to abide by relevant laboratory safety procedures.	
Waterproof Warning It indicates that infiltration may cause damage to the instrument in the adjacent area of this sign. Do not place liquid in this area.	
Mechanical parts Warning It indicates a potential danger in case of touching the moving mechanical parts in the adjacent area of this sign.	

Preventive measures on operation

In order to correctly use Automatic Hematology Analyzer, pay attention to the following matters:

Notes to ensure accurate test results

- Do not touch the instrument test module cover plate while the instrument is operating, otherwise, the test results will be incorrect.
- Please replace the maintenance parts timely; otherwise, the test results may be incorrect.
- Clean the sampling needle termly to keep the instrument surface free of dirt, otherwise, there will be instrument fault or the test results may be incorrect.
- Be sure not to use expired reagent, calibrators and control. Use expired products will cause test error and the test results will be incorrect.
- Do not place samples, calibrators or control in the sample tube for a long time, otherwise it will be condensed because of evaporation, and it cannot be accurately analyzed and lead to an inaccurate results.
- Please use standard sample tube, if not, the test results may be incorrect because of the inaccurate absorption quantity of the samples.
- Do not adding reagent, otherwise it will cause cross-contamination, resulting in the inaccurate results.
- During test, please be sure not to open the cover of the reagent cabinet, otherwise, there will be instrument fault or the test result will be incorrect.
- Pipeline filling must be made before the start of every day, otherwise it will cause the inaccurate results.

Notices for prevention of instrument damage

- Please use standard sample tube; if not, the sampling system may be damaged.
- Overflow of liquid from the sample needle wash tank might cause instrument fault such as damages of circuit board or subassembly.
- Be sure not to put the sample or reagent on the platform of the instrument. Splash of sample or reagent on the instrument might cause instrument fault such as damages of circuit board or subassembly.

Notices on instrument power on

- Switching power supply during instrument maintenance is not allowed because it may cause personal injury or instrument damage.

Notices on test start

- Be sure maintenance of the instrument is not going on, and then click the start button in the right window of the software. If test starts during instrument maintenance, parts or tools used for maintenance may touch the running mechanism, which will cause instrument damage.

Data backup

- In case of instrument fault or faulty operation, the test results will disappear. For security purpose, a regular backup update of the test results is recommended.

Measures against computer viruses

- If there is a sudden corruption of program / data, or unexpected situation pops up in the operation or on the screen, it is possible that your computer is infected by viruses.
- Computer virus refers to malicious programs that break into your computer to cause data exception or corruption.
- The program designed to protect against computer viruses is called antivirus programs.
- Possible causes of the virus infection:
  Program with virus was downloaded in communication process.
  USB drive or other storage medium infected with virus are mistakenly used.



Notices:

- Once a computer is infected with virus, other computers are likely to be infected by means of communication or storage medium.
- Be sure not to use programs or storage medium with suspected virus.
- Before use of mobile storage medium, please scan with the antivirus program.

Interruption of power supply

- In case of interruption of power supply or thunder strike, the operation unit of the instrument may break down, and the software system, application software and data may get damaged.

Control unit

- During the access to a hard disk or other storage medium, if there is a sudden loss of power in the control unit, the data/ software stored there may get damaged. Please refer to relevant chapters and sections in the user manual for correct method of switching off the control unit.
- Be sure not to install other programs into the control unit or change the unit settings.
- Do not remove the mobile medium when the indicator of the mobile medium is on. Otherwise, the instrument or the mobile medium will get damaged.
- Be sure not to put magnetic assays near the display. Otherwise, these assays will get damaged.

Notices for prevention of electromagnetic wave interference

- Be sure not to place the instrument near the source of electromagnetic wave interference, otherwise, there can be interference and the test result can be

affected, even to cause instrument fault.

The name and content of toxic or harmful substances or elements

The name and content of the toxic and harmful substances or elements in this product are shown in Table 1.

Table 1： The name and content of toxic and harmful substances or elements in the product

component		Toxic or harmful substance or element					
		Plumbum (Pb)	Hydrargyrum (Hg)	Cadmium (Cd)	Six valent chromium (Cr(VD)	Polybrominated biphenyls(PBB)	Polybrominated diphenyl ethers (PBDE)
Automated Hematology Analyzer	Circuit board	O	o	O	O	O	O
	Electric wire	o	o	o	o	O	O
	Mechanical parts	o	o	o	o	O	o
	Liquid path system	o	o	o	o	o	o
	Machine frame	o	o	o	o	o	o
	Outer shell	o	o	o	o	o	o
	PC (Control computer)	o	o	o	o	o	o
	Monitor	o	o	o	o	o	o
	Printer	o	o	o	o	o	o
	ID unit (barcode reading unit)	o	o	o	o	o	o

O： It is indicated that the content of the toxic and harmful substances in all the homogeneous materials of the component is below the limit specified in the SJ/T 11363-2006 mark.

x： It is indicated that the content of the toxic and harmful substances at least in a certain homogeneous material of the component is beyond the limit requirements specified in the SJ/T 11363-2006 mark.

Chapter 1
Overview of manual

1.1 Overview

This chapter describes how to use the Automatic Hematology Analyzer manual. This manual comes with instrument, and provided with a detailed description of the use, function and use method of the analyzer. Before using the analyzer, please read and understand this manual carefully to ensure proper use of the analyzer, to perform its best performance and to ensure operator safety.



Important
Please use the instrument according to the manual strictly.

1.2 Application Range

This manual applies to medical inspection professionals or the trained doctors, nurse and laboratory technician. Used for:

- Understanding the instrument hardware and software
- Setting the system parameters
- Routine analysis of the sample
- Performing system maintenance and troubleshooting

1.3 Common operate

Name	Operate
click	Left click on the corresponding position or button, sometimes need to double-click the button.
input	Click on the input or edit box, the cursor appears in the input box, use the keyboard to enter the required information (letters, characters, numbers, symbols, etc).
delete	Click the left button of mouse (or use the ↑ ↓ ← → of keyboard) to locate the cursor at the delete place, backspace key to delete the cursor before the information, delete key to delete the cursor information.
scroll bar dragging	The left mouse button click the arrow on both sides of the scroll bar to perform the page flip operation; Or the mouse to select the scroll bar, move the mouse to flip the page.
pull-down list choosing	Mouse click drop-down box down arrow, pop-up drop-down list, mouse click after the move or click the keyboard ↑ ↓ key to select the Enter key to confirm.

1.4 Symbol description

The following symbols are used in the manual, the symbolic meanings are as follows:

symbols	meaning
	The operator is prompted to operate according to the instructions in the symbol otherwise it may result in personal injury.
	The operator is prompted to operate according to the instructions in the symbol otherwise it may result in analyzer failure, damage or influence test results.
	The operator is prompted to operate according to the instructions in the symbol, emphasize the important information in the operation step or the content that requires special attention from the operator.
	The operator is prompted to operate according to the instructions in the symbol, otherwise there is the danger of potential biocontagiousness.
	Do not open doors or lids in normal work, if you need open the door or the lid, it must be done by the person who has been trained by the relevant training.
	The operator is prompted to operate according to the instructions in the symbol, otherwise, it may cause damage to the components of the product, which may affect the test result or bring personal injury.

In the analyzer, reagent packaging, quality control products or calibration products may show the following signs:

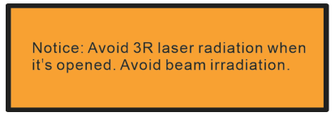
signs	Meaning
	Warning signs
	High pressure warning
	Avoid light warning
	Biological pollution

	High temperature warning
	Protective Grounding
	Alternating current
	Manufacturer
	Production date
	Refer to the instructions
	The serial number
	It is only used for in vitro diagnosis
	Expiry date
	Measuring permission
	Storage temperature
	Batch number
	Antistatic labeling

1.5 Laser Introduction

Laser type	Wavelength	Power	Maximum output	Horizontal transmission Angle	Vertical Angle of transmission
List 3B	635nm	3.5mW	5mW	8°	31°

You might see the following laser warning labels outside or inside of the instrument, please follow the instructions to prevent from laser damage!



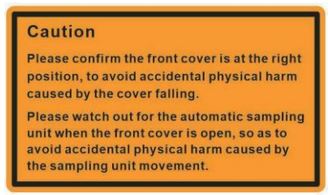
This label must be pasted on the optical component laser shield box.



This label must be pasted on the fuselage shell.

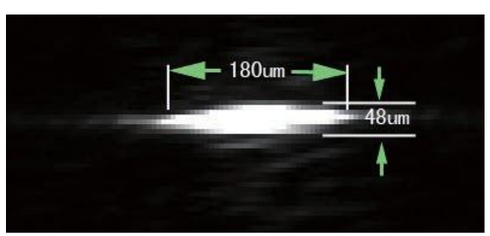


This label must be pasted on the optical component laser shield box inside the machine.



This label must be pasted on the instrument cover. Laser focus

spot size:



1.6 Warning



- This system only can be used by the professional medical examiners, doctors or lab operators who be trained by our company or distributors of our company.
- If the hospital or institution which is responsible for the analyzer cannot provide a good maintenance plan, it may lead to the failure of the normal analyzer and endanger the human health.
- Please ensure to use the analyzer under the operating instructions. If the operation condition exceed the user manual regulated, the analyzer may not function normally, and the test results are unreliable and may damage the analyzer components and damage the human safety.
- The analyzer must be used in good grounding conditions.
- The analyzer must use the specified fuse.
- Please confirm the voltage meets the requirements.
- Do not place the instrument near the electromagnetic interference
- source, otherwise it will be interfered and will affect the test result and even cause the instrument malfunction.
- The analyzer should not be used in flammable and explosive environment to avoid the danger.
- Do not move the analyzer without authorization. If you need to move, please contact the service department of this company.
- Reagents can irritate eyes, skin and mucous membranes. Operators should comply with laboratory safety regulations and wear personal protective equipment when using contact reagents related items (such as Lab protective coats, gloves etc.).
  If the reagent contact with the skin, use water to clean it immediately, or ask for doctors if needed.
- The discharge of the reagents, waste liquid and waste sample should obey the regulations in the corresponding country, which related to the
- discharge and disposal of potential inspection materials for the reagent and waste liquid.
- Due to the sampling needle is too thin, and the potential inspections will be caused by the calibrators, quality controls and samples. Please pay attention during operation.
- The disposal of the waste calibrators and the package of quality controls should obey the regulations in the corresponding country, which related to the discharge and disposal of potential inspection materials for the
- reagent and waste liquid.
- Only the engineers and operators approved by our company can operate and maintain the analyzer, otherwise the harmful radiation hurt can be caused.
- It is forbidden to open the laser shield during normal working. Please contact our company as soon as possible if the protective cover has loose or large gap.

1.7 Notice

Important

- The reader of this instruction is the following professional staffs which in the following clinical department:

1. Staffs who perform the daily system operation;
2. Staffs who perforin the system maintenance and troubleshooting; Staffs
3. who learn system operations.

- Please follow the following instructions strictly.
- The analyzer is used for clinical screening. When the doctor making decisions according to the analytical results, clinic test results or other test results should be considered simultaneously.
- Please refer to the reagent instructions for the storage and use of reagents.
- Check the residual amount of the reagent daily before using the analyzer, and estimate whether or not it is sufficient for the specimen of the day. If there are any shortages for the reagent, it should be prepared in time.
- After replacement of any reagent, a background check should be carried out to ensure that the measured values are within the allowable range of the background, and the specimen test can only be carried out after being qualified.
- The reagent should be restored from storage temperature to room temperature.
- The instrument might be damaged by the staff who did not authorized or trained by our company during open box or installation process, please do not open box or installation analyzer when the staff whom authorized by our company are not here.
- Please do not place reagents or other liquids on the top of the analyzer. There is no order between the power for the software of Mispa HX50 and the power of the analyzer, and one of them can be chosen randomly.
- The operator should use a clean EDTA-K2 anticoagulation vacuum blood collection tube, silicified glass tubes, plastic test tubes, centrifuge tubes and silicon-borated glass capillary.
- The amount of sample is not less than 1 mL in manual whole-blood mode. The samples which use to white blood cell classification or platelet count should be store at ambient temperature and analyzed within 4 hours after collection.
- If the results of the classification of platelets, the average red blood cell volume or leukocyte classification are not required, it can be stored for 24 hours at 2℃ ~ 8℃.And after refrigeration, samples should be placed at ambient temperature for at least 30 minutes before analysis.
- The sample should be remixed after stewing for a while.
- The peripheral blood and the diluent need to be placed for3minutes before being tested.
- Please ensure that the capillary blood sample tested within 30 minutes after dilution.
- The laboratory should evaluate the stability of the test results in the pre-dilution mode which based on the number of samples, the sample collection method and the technical level.
- The "***" represents the sample number entered in storage.
- Please do not touch the sampling needle during removing the sample tube, and prevent from the collision of the sampling needle or pop the blood in the tube.
- After finishing analysis of a sample, the status bar display sample number will be automatically increasing +1, so as long as the work list is registered in the order of registration, the manual smaple number will not be needed to intervene when the test is processing. In order to guarantee the reliability and stability of the analyzer, Be sure to execute the shutdown procedure.
- If users want to cancel shutdown program, please click the ,cancel" button in the shutdown dialog box and then back to the measurement ready status.
- After switching off the power, we can still use the Mispa HX50 software work sheet, result view, result printing and other functions that have no interaction with the host.
- The results of the batch selection are displayed with high brightness, Result display column only display the sample arrow indicated on the left list.
- The sample number start with QC is the quality control data and not allowed to be modified.

  The audit status is shown in the audit column with the "S" tag.
- The specimen number is not allowed to be modified to start with "QC-
- ″ or ″qc- ‘to avoid confusion with quality control data.
- The number in the dialog box is the number of the results selected by the user batch selection.

  Before print out the report, please confirm that the printer installation are connection are in good status, and papers are sufficient.
- The files are automatically produced by the system, and users can rename them as needed.

  Save the type as the only format CSV.
- When the sample performs the CBC+DIFF test mode, the actual display of the WBC histogram is the BASO classification figure.
- Double click on the graph to enlarge the display.

  The reviewed results are shown in blue background in the review status bar, and the unreviewed results are shown in white background.
- The sample number start with "QC is the quality control data, which cannot be printed out in browser or manager. It must be output in quality control print format on QC chart interface.
- Only login users with administrator privileges can perform the calibration. The user shall use the quality control products and calibrations designated by the company, and strictly in accordance with the calibration, quality control and reagents specified storage environment for storage.
- Precision test method: In whole blood test mode, test the middle level quality control 10 times continuously, export the results to CSV format and use Excel to count CV values.
- Only prescribed quality control products can be used. The quality control products are used to judge the analysis status of the analyzer.

  When the batch number is not entered, the confirm button is not enabled.

- Click enter to modify the registered document number, the quality control materials can not be modified
- When the selected document is blank and the batch number information is not entered, the delete button is not enabled.
- Confirmation dialog box "Totally [1] ", the number represents the currently selected number of documents to be deleted.
- Incorrect maintenance can damage the analyzer. The operator must perform maintenance according to the specification.
- If there is no clear problem in the instruction manual, please contact the service department of our company to give the maintenance advice by professional staff.
- The analyzer must be maintained using the parts supplied by our company. If you have any questions, please contact our service department.
- This manual is not a maintenance manual. It only provides the measures that the operator should take when the analyzer fails or alarms.

1.8 Biological risk

- Samples, calibrators, control and waste liquid have the risk of potential bio infectious. Please observe the laboratory safety operation rules and wear personal protective equipment (such as laboratory protective clothing, gloves, etc.)
- Do not contact the patient's blood samples directly.
- All parts of the analyzer are potentially bio infectious, and safety precautions should be taken during operation and maintenance.



Chapter 2 System Introduction

2.1 Overview

Automatic Hematology Analyzer is an in vitro diagnostic equipment, and it is used together with supporting reagents, calibrators, and quality control material. It is used in medical laboratory to quantitatively analyze blood cells and conduct 5-part differentiation on WBC.

2.2 Scope and Contraindications

The analyzer is suitable for testing human blood samples, conducting qualitative and quantitative analysis of the visible components of blood, and providing relevant information.

Important	■ The analyzer is the instrument for clinical screening. Doctors should take into account of the clinical examination results or other test results when making judgments based on the analysis results. ■ Contraindications: This analyzer is used in clinical anticoagulant testing, and samples are required to have no clotting and no hemolysis.

The quantitative analysis of following 24 report parameters and 4 research parameters, three histograms, and one scattergram are provided.

-	Name	Abbreviation	CBC	CBC+ DIFF
CBC Parameters	White Blood Cell Count	WBC	※	※
	Red Blood Cell Count	RBC	※	※
	Hemoglobin	HGB	※	※
	Mean Corpuscular Volume	MCV	※	※
	Mean Corpuscular Hemoglobin	MCH	※	※
	Mean Corpuscular Hemoglobin Concentration	MCHC	※	※
	RBC Distribution Width-Coefficient of Variation	RDW-CV	※	※
	RBC Distribution Width-Standard Deviation	RDW-SD	※	※
	Hematocrit	HCT	※	※
	Platelet Count	PLT	※	※
	Mean Platelet Volume	MPV	※	※
	Platelet Distribution Width	PDW	※	※
	Platelet Hematocrit	PCT	※	※
	Large platelet ratio	P-LCR	※	※
DIFF Parameters	Basophilic Granulocyte Count	BASO#	/	※
	Basophils Percentage	BASO%	/	※
	Neutrophile Granulocyte Count	NEUT#	/	※
	Neutrophile Granulocyte Percentage	NEUT%	/	※
	Eosinophilic Granulocyte Count	EO#	/	※
	Eosinophilic Granulocyte Percentage	EO%	/	※
	Lymphocyte Count	LYMPH#	/	※
	Lymphocyte Percentage	LYMPH%	/	※
	Monocyte Count	MONO#	/	※
	Monocyte Percentage	MONO%	/	※
	Other Cells Count (Research)	OTHER#	/	/
	Other Cells Percentage (Research)	OTHER%	/	/
	Immature Granulocyte Count (Research)	IG#	/	/
	Immature Granulocyte Percentage (Research)	IG%	/	/
Histograms	White Blood Cell Histogram	WBC Histogram	※	※
	Red Blood Cell Histogram	RBC Histogram	※	※
	Platelet Histogram	PLT Histogram	※	※
Scattergram	Differentiation Scattergram	DIFF scattergram	/	※

"※" indicates that this parameter is tested under this mode;"/" indicates that this parameter is not provided under this mode.

2.3 Structure and Components of the Product

The analyzer is mainly composed by the mainframe and accessories.

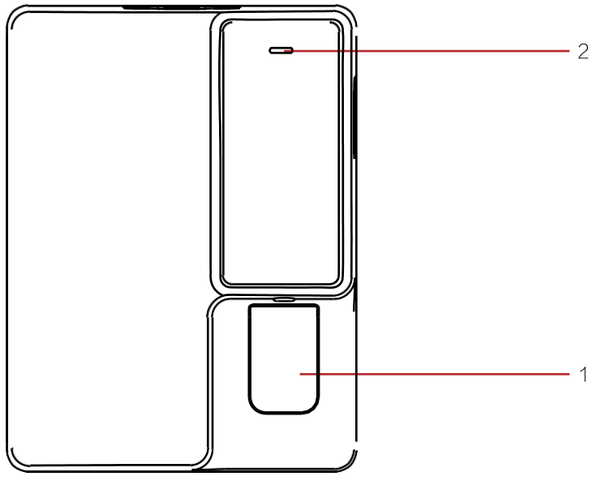


Figure 2.3-1

1	Sampling Start Button
2	Status Indicator Light

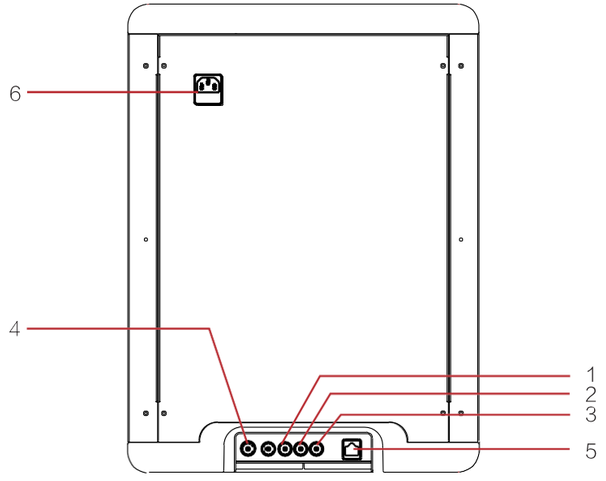


Figure 2.3-2

1	LH-5 Lysing Reagent Interface	4	Liquid Waste Interface
2	LD-5 Lysing Reagent Interface	5	Network Line Interface
3	GD-5 Diluent Interface	6	Power Cord Connector
7	Sensor interface		

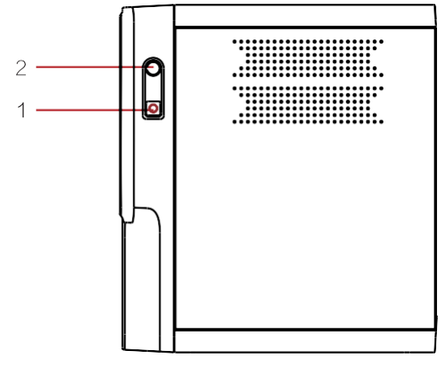


Figure 2.3-3

1	Alarm elimination button	2	Power Switch

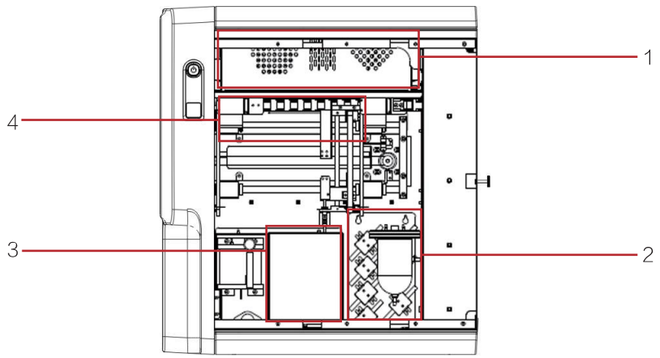


Figure 2.3-4

1	power	3	WBC, RBC Detector Block
2	Liquid Waste Chamber	4	Sample Collection Unit

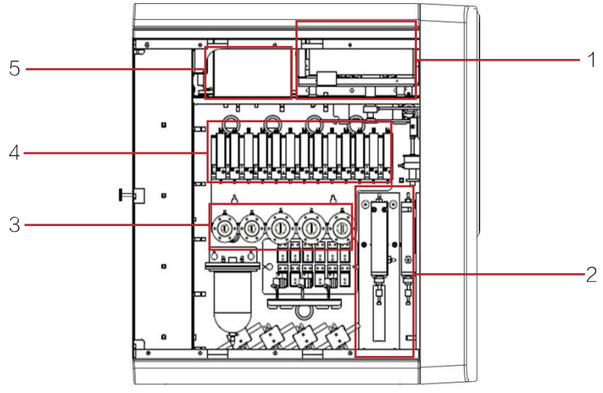


Figure 2.3-5

1	Optical System
2	Syringe
3	Dosing Pump Device
4	Liquid Valve
5	Air Source

2.3.1 Status Indicator Light

Status indicator light is located at the front of the analyzer. It displays whether the analyzer is in the Ready or Not Ready state currently.

2.3.2 Aspiration Pipette

Aspiration pipette is located in front of the analyzer. It accurately draws quantitative blood samples.

Notices:



- The sample needle is not exposed when the instrument is turned off and the exposure during normal use is unavoidable. It must be carried out by personnel who have been approved by relevant training. Otherwise, the analyzer may not run normally, the measurement result may be unreliable, and may damage the analyzer components and cause personal injury.

2.3.3 Sampling start button

The sampling start button located on the front of the analyzer, triggering the implementation of the corresponding timing analyzer.

2.3.4 Instrument interface

- Power Interface
  For the insertion of the power cord which is to be connected with the network.
- Reagent Waste Interface
  Connect analyzer with various supporting reagents and waste bucket by tubes.
- Network Line Interface
  Built-in 100M network card, compatible with TCP / IP protocol. Insert the cable to communicate with supporting software of analyzer, issue appropriate command and upload analysis data.

Important

- The computer connected with the instrument must meet the relevant standards.

2.3.5 Power switch

Power switch is located on the right side of the analyzer. It is used to connect and disconnect the power.

2.3.6 A key alarm button to eliminate

One-touch alarm elimination button located on the right side of the analyzer for one-click elimination of alarm and recovery of instrument status to ready.

2.4 Interface

2.4.1 Main interface

Double click the icon on the desktop, and the main interface as Figure 2.4.1-1 shown will appear on the screen after startup. Double click the icon to enter the corresponding function interface.

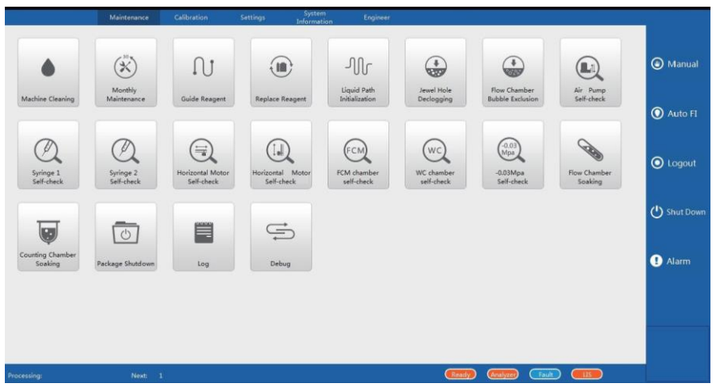


Figure 2.4.1-1

2.4.2 Main Function

By clicking the corresponding button, you can enter the corresponding function interface to perform different functions.

Button	**Function**
Work station	Register work list sample View the sample analysis results Review, review, print sample result
History data	Different ways to review the analyzed sample results Query the analyzed sample results
Detailed browser	Check the results of an analyzed sample in detail
QC	Register or view quality control documents View quality control history with quality control chart Perform quality control analysis
Basic data	Register, modify, check the information of the patient, doctor, department and ward building
Maintenance	Perform maintenance operations
Setting	Set the related functions
System	Perform the operation of calibration, dormancy, switch user, logon audit, shutdown, view instrumentation information.

2.4.3 Functional Interface

- Menu Interface
  Provides 5 menu buttons to click and unfold.
- Shortcuts Area
  Click to perform related functions

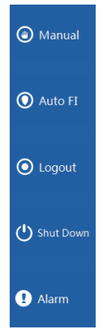


Figure 2.4.3-1

- Instrument Status Interface Displays current status of instrument

→Status-Analyzer

Meaning: network status of software and instrument, divided into unconnected (red) and connected (blue).

→ Status-LIS

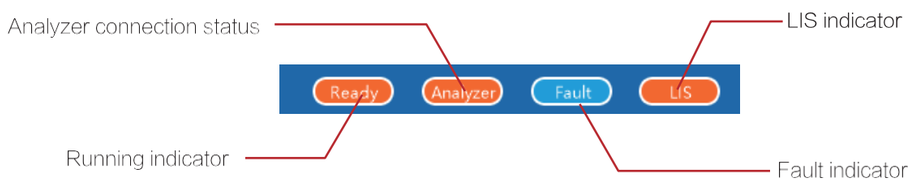
Meaning: The connection state between the program and LIS, divided into unconnected (red) and connected (blue).

→ Status-fault

Meaning: The current fault status of the analyzer, divided into failure (red), normal (blue).

→ Status-operation

Meaning: The operating status of the analyzer, divided into unready (red), ready (blue) and running (green).



Two buttons including manual sampling and automatic fluid injection, it can perform related operations.

→ Manual sampling

When the analyzer is ready and without alarm, click the manual sampling button to perform manual sample editing and saving, and then click
the corresponding button on the instrument to perform manual sample measurement.

→ Automatic liquild injection

When the analyzer is ready and without alarm, click the automatic liquild injection button and the instrument can be operated automatically.

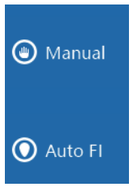


Figure 2.4.3-3

The following areas are used for related sample information, including:

→ Test information area

■ Show the current sample and the next sample number to be tested.

→ Analysis setting area

■ Click the boot mode to pop up the dialog box below.

■ Manual sampling:

Figure 2.4.3-4

→ Fault information area

Display fault information, if the number of faults is too large, it can be displayed by rolling. Double click on fault information can enter the fault detailed interface. Click the button ''Clear Alarm" to eliminate the fault, click the button “Stop Beep” to eliminate the alarm sound when the current fault occurs.

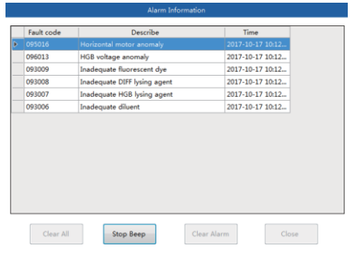


Figure 2.4.3-5

2.5 Samples description

2.5.1 Sample classification description

Sample sampling method is only manual sampling; quality control sample, calibration sample, and background checking sample are all classified into manual sample type.

According to the way of sample generation (or matching mode), it is divided into non scanning generation and scanning production samples. Non scanning generation is also divided into work sheet samples, automatically generated samples (automatic increments). The scanning mode requires the apparatus to configure the bar code scanning module.

The worksheet sample can be registered before the test, and the untested worksheet sample number is not allowed to repeat.

Every day the sample number starts from 1", and if the work sheet has already registered the sample "1", it begins to measure from the work sheet. Automatically generated samples are generated automatically during the measurement process (Before the measurement, the user can set the number from which to start). The last sample is +1 based on the previous one, for example, the former one is ST, and the latter one is S2.

The sample number generated by the scan is the barcode number, and the barcode is scanned by the analyzer during the measurement. Scanning match mode requires the support of analyzer scan module.

2.6 Reagents, Calibrators and Control

Analyzer, reagents, calibrators and control constitute a system together, and must be used as a whole to ensure overall performance.

Reagents must be designated by Agappe, otherwise the reliability and accuracy of analytical results cannot be ensured, and may cause damage to the analyzer. "Reagent" in this manual all refer to supporting reagents of the analyzer.

Each reagent J s package integrity and validity must be checked before usage. Make sure that no package is damp or leaking, and the reagent must be used within the period of validity.

Important

■ Please refer to instructions for reagent storage and usage.

■ Check reagent remaining before using analyzer, and estimate whether reagent remaining is enough for the day, if not please prepare in advance timely.

■ Background should be checked after changing any reagent to ensure that the measured value is within the allowable range. The samples only can be tested after passing background test.

■ Reagents should recover from storage temperature to ambient temperature when being used.

2.6.1 Reagent

- Automatic Hematology Analyzer GD-5 Diluent A
- Diluent is an isotonic solution that has a specific conductivity, provides a stable environment for the blood count and conducts pipe cleaning after completion of the test.
- Automatic Hematology Analyzer LH-5 Lysing Reagent A
  Dissolve erythrocytes to conduct HGB measurement.
- Automatic Hematology Analyzer LD-5 Lysing Reagent A
  Dissolve erythrocytes and classify the leukocytes.
- Automatic Hematology Analyzer DD-5 Fluorescent Dye A
  Stains nucleic acid of leukocytes and classify the leukocytes.
- Automatic Hematology Analyzer CC-5 Cell Clean A
  It contains detergent which can remove stains in the tubes. It is used for routine instrument cleaning and maintenance.

2.6.2 Control and Calibrators

Control and calibrator are used for quality control and analyzer calibration.

''Control material' and "Calibrator" mentioned in this manual refer to Agappe designated control material and calibrators, which must be purchased from Agappe or Agappe agents.



Chapter 3 Design Principles

3.1 Overview

The analyzer uses electrical impedance method to measure the number of red blood cells and platelets; uses colorimetric method to measure the concentration of hemoglobin; and uses semiconductor laser flow cytometry technique to count WBC and classify five-diff WBC. And calculate the remaining parameters according to the corresponding equations.

3.2 Samples Extraction

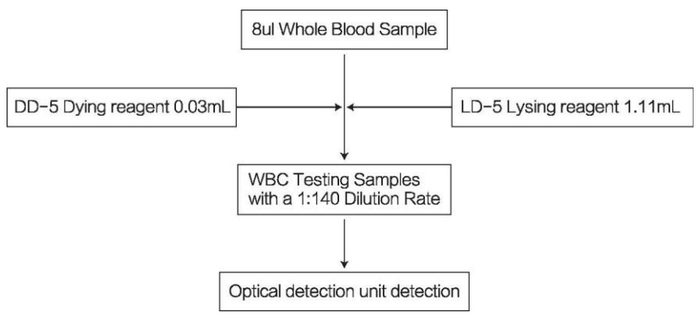
Analyzer provides an open sampling method.
Open sampling supports whole blood and pre-dilution mode.
Under open whole blood mode, the analyzer will draw 20 u l of whole blood samples.
Under open pre-dilution mode, the operator will first mix 20 u l peripheral blood sample with 200 u l dilution to form a sample of 1:11 dilution outside the analyzer.

3.3 Sample Dilution

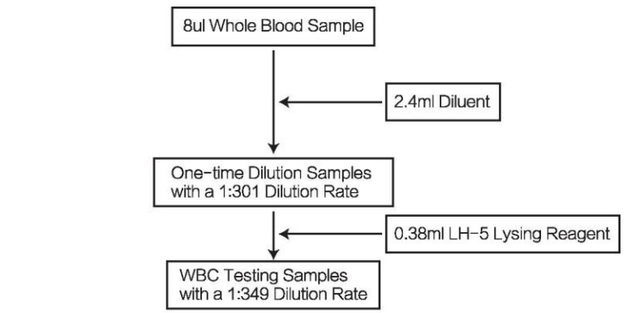
After blood samples are collected, blood will be distributed and diluted according to different measurement parameters, and then be used for leukocyte differentiation and counting, hemoglobin measurement, and erythrocyte counting and measurement.
Based on the different options of the user, the analyzer provides two operation modes: whole blood mode and pre-dilution mode.

3.3.1 Whole Blood Mode

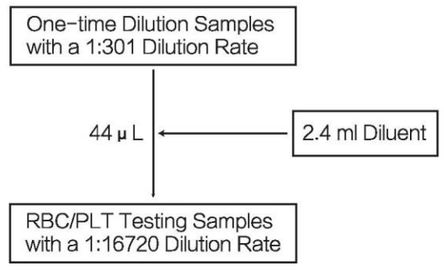
■ WBC Differentiation Dilution Process



■ WBC Counting / HGB Dilution Process



■ RBC/PLT Dilution Process



3.4 WBC Measurements

3.4.1 WBC Differentiation (Semiconductor Laser Flow Cytometry)

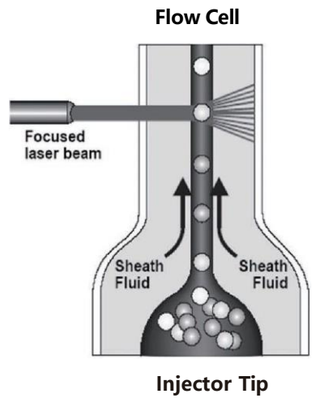


Figure 3.4.1-1

When certain amount of blood cells are drawn and reacted with specific reagent, certain amount of blood samples will be injected through the nozzle

into the conical flow chamber which is filled with diluent. With the surrounding of sheath fluid formed by diluent, the cells will cross the central flow chamber in a single row. When the blood cells which are floated in the sheath fluid pass through the laser detection region, they will be exposed to a laser beam and produce scattered lights, whose property is associated with the cell size, cell membrane and refractive index of the internal cell structure. The forward scattered light (FSC) reflects the size of the cells, the side scattered light (SSC) reflects the cells' fine internal structure and particulate matters. The photodiode receives these scattered light signals and converts them to electric pulses. According to the collected electrical pulse data, the two-dimensional distribution diagram of blood cell size and internal information can be obtained, which is called scattergram. In Figure 3.4.1-2, the abscissa reflects the information of the complexity of cells inside and the ordinate reflects the volume of the cell.

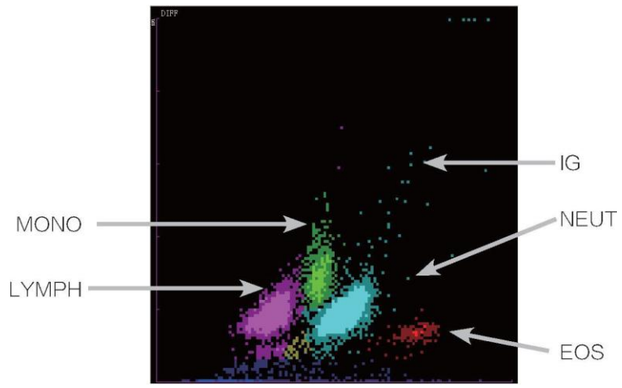


Figure 3.4.1-2

From DIFF channel, percentage of LYMPH, MONO, EOS, NEUT and IG of the WBC total count can be obtained.

3.4.2 WBC Counting (Coulter Principle)

When certain amount of blood cells react with a specific reagent, RBC and WBC excluding BASO are dissolved. With the reaction liquid driven by constant negative pressure, the cells will one by one pass through a small hole applied with constant current source. The number and height of pulses reflect the number and volume size of the cells. Principles are shown in Figure 3.4.2-1.

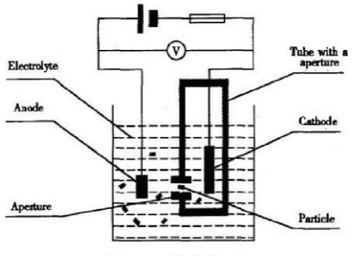


Figure 3.4.2-1

From the WBC histogram, WBC count, BASO count and the percentage of BASO of total amount of WBC can be obtained. Histogram is shown as Figure *3.42-2.*

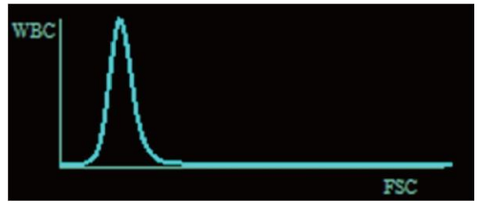


Figure 3.4.2-2

3.4.3 WBC Parameters

By analyzing the WBC histogram, the analyzer can compute White Blood Cell (WBC) count and basophile count (BASO#), and further get the basophile percent (BASO%). It can compute the neutrophils percent (NEUT%), the lymphocyte percent (LYMPH%), the monocyte percent (MONO%), the eosinophile percent (EO%) and the immature granulocytes percent (IG%) by analyzing the scattergrams and its area of neutrophile, lymphocyte, monocyte, eosinophile and immature granulocytes. Computing with the WBC count obtained from the WBC histogram, you can get the neutrophile count (NEUT #), the lymphocyte count (LYMPH#), the monocyte count (MONO#), the eosinophile count (EO#) ,the basophile count (BASO#) and the immature granulocytes count (IG #). The units of the cell count are 109/ L.

■ WBC

WBC= Total number of particles on the WBC histogram except in the ghost

■ NEUT%

$$NEUT\% = \frac{\text{Particles that fall in NEUT area from scattergram channel}}{\text{Total number of particles in the scattergram except in the ghost area}} \times 100\%$$

■ NEUT#

$$NEUT\# = NEUT\% \times WBC$$

■ LYMPH #

$$LYMPH\% = \frac{\text{Particles that fall in LYMPH area from scattergram channel}}{\text{Total number of particles in the scattergram except in the ghost area}} \times 100\%$$

■ LYMPH#

$$LYMPH\# = LYMPH\% \times WBC$$

■ MONO%

$$MONO\% = \frac{\text{Particles that fall in MONO area from scattergram channel}}{\text{Total number of particles in the scattergram except in the ghost area}} \times 100\%$$

■ MONO#

$$MONO\#=MONO\% \times WBC$$

■ EO%

$$EO\%=\frac{\text{Particles that fall in EO area from scattergram channel}}{\text{Total number of particles in the scattergram except in the ghost area}} \times 100\%$$

■ EO#

$$EO\#=EO\% \times WBC$$

■ BASO%

$$BASO\%=\frac{BASO\#}{WBC} \times 100\%$$

■ BASO#

BASO#=Particle numbers that fall in the BASO area from WBC histogram

■ IG%

$$IG\%=\frac{\text{Particles that fall in IG area from scattergram channel}}{\text{Total number of particles in the scattergram except in the ghost area}} \times 100\%$$

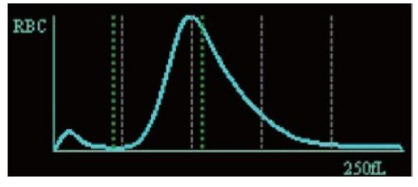
■ IG#

IG#=IG%×WBC

3.5 RBC/PLT Measurements

3.5.1 Coulter Principle

The analyzer uses the Coulter Principle to count the RBC/PLT. A certain amount of blood cells are injected into RBC reaction chamber after secondary dilution. There is a small hole called detect hole in the RBC reaction chamber. The small hole is connected with positive and negative electrodes on the side and is charged with constant current source. Since the cell is a poor conductor, the resistance between the electrodes will change when the diluted cells pass through the hole, thereby forming a pulse signal with the similar size of the cell volume at both sides of the electrodes. When cells pass through the hole continuously, a series of pulses will be formed. Pulse number is consistent with the cell number, and the pulse height is related to the size of the cell. Principle is shown in Figure 3.5.1-1.



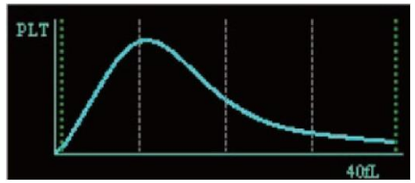


Figure 3.5.1-1

3.5.2 RBC Parameters

■ RBC Count
The RBC count is obtained by direct measuring corresponding electrical pulses numbers. Unit is 10^{12}/L.

■ Hematocrit (Unit %)

■ According to the histocyte distribution histogram, and then detected Hematocrit (HCT), and the unit is %.

■ Mean Corpuscular Volume

$$MCV=\frac{HCT \times 10}{RBC}$$

■ Mean Corpuscular Hemoglobin

$$MCH=\frac{HGB}{RBC}$$

■ Mean Corpuscular Hemoglobin Concentration

$$MCHC=\frac{HGB}{HCT}X100$$

■ RBC Distribution Width Coefficient of Variation
RBC Distribution Width Coefficient of Variation (RDW-CV) is a coefficient of variation which is expressed in percentage and obtained from the RBC histogram.

■ RBC Distribution Width Standard Deviation
RBC Distribution Width Standard (RDW-SD) is the histogram width at 20%of the peak height on the histogram. Unit is fL. It is shown in Figure 3.5.2-1

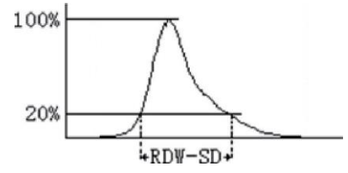


Figure 3.5.2-1

3.5.3 PLT Parameters

■ PLT Count
The PLT count is obtained by direct measuring corresponding electrical pulses. Unit of 10^{9}/L.

- Mean Platelet Volume

  Compute Mean Platelet Volume(MPV)according to PLT histogram.Unit is fL.

- Platelet Distribution Width

- Platelet Distribution Width(PDW)is a geometric standard deviation which is obtained from the PLT histogram.

  Platelet Hematocrit

  The following equation can be used to compute the platelet hematocrit.(PCT) Unit is %.

$$PCT = \frac{PLT x MPV}{10000}$$

3.6 HGB Measurement

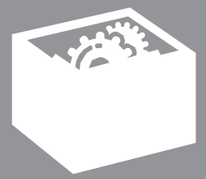
3.6.1 Colorimetry

In the colorimetric pool, after the diluted sample is added to the lysing reagent, red blood cells will dissolve and release hemoglobin, which unites with lysing reagent and becomes hemoglobin compounds. At one end of the colorimetric pool, illuminate the hemoglobin compounds with LED light emitted from a monochromatic light tube with 525nm wavelength; the other end receives transmission lights through phototube ,and converts them to voltage signal after the optical intensity signal is amplified. By comparing it with the voltage obtained from the transmission light before the samples were added to the colorimetric pool(only diluents before),you can get the sample's hemoglobin concentration.

3.6.2 HGB Parameters

HGB is calculated by using the following equation, unit is g/L.

$$HGB = Constant \times Ln\left(\frac{\text{Background Transmission Light}}{\text{Sample Transmission Light}}\right)$$



Chapter 4 System Installation

4.1 Overview

Automatic hematology analyzer has been checked strictly before leaving the factory and is packaged properly to avoid impact in the transportation. Please check packaging integrity carefully upon delivery. If there is any damage to the packaging, please inform the Post-sales Department of AGAPPE or its local representative immediately.

Important	■ Unpacking or installation by personnel unauthorized or untrained by AGAPPE may cause damage to the analyzer, so please do not open the package or install the analyzer without the presence of AGAPPE authorized personnel. ■ If the device is not used over a long period of time, please execute the package shutdown program to empty reagents inside the device and wash it with deionized water. Device storage conditions should comply with the environmental requirements.

4.2 Installation Requirements

Make sure that the space, power, and environment meets the following requirements before installing the analyzer.

4.2.1 Space Requirements

In order to meet the equipment s maintenance space demands, and to ensure heat dissipation during usage and reagent pipe at the back of analyzer not be squeezed, the installation of the analyzer must meet the following space requirements:

- Reserved space between both side-doors and the walls should be > 100 cm;
- Reserved space between rear panel and the wall should be > 50 cm;
- Make sure that there is enough space for dilution, lysing reagent and waste barrels on the table or under the analyzer.
- Do not place the analyzer in a position where the switch is difficult to disconnect.

4.2.2 Power Requirements

Voltage	Power	Fuse Specifications	Frequency
AC 100 - 220V	≤ 240VA	250V T5A	50Hz/60Hz

Warning

- Analyzer must be used in well-grounded conditions.
- Device must use the fuse of specified standard.
- Make sure that the voltage meets the requirements.

4.2.3 Environmental Requirements

- Environmental temperature: 15℃ ~ 30℃;
- Normal operating humidity range: 30%~85%;
- Atmospheric pressure:70kPa ~ 106kPa;
- Installation environment should be as free of dust, mechanical vibration, noise and strong power interference as possible;
- Do not place the device near strong electromagnetic interference sources, so as not to affect the normal operation of the device;
- Do not go near the brush-type motors, fluorescent lights flicker and the constantly switching electrical contact device;
- Device should avoid direct sunlight or being placed near a heat source;
- Keep the device ventilated;

Warning

- Do not use analyzer in flammable or explosive environments, so as to avoid danger.
- If the instrument is to be moved to a different location at a

4.2.4 EMC Requirements

The instrument need good EMC environment, users should notice the following situation:
Keep this instrument away from the high electromagnetic radiation, such as, dental drill, pacemaker and welding equipment etc.
Keep this instrument away from the high power devices which start frequently, such as, refrigerater and centrifuge etc.
Keep this instrument away from any magnetic field, such as radiology equipment.
Do not connect more devices on the wiring board for connecting this instrument.
During the proceeding of this experiment, do not start other devices on the wiring board for connecting this instrument.
Please ensure that the power outlet of this instrument is securely grounded, and do not use the non-standard power cord.
This instrument passed the electromagnetic compatibility testing, which satisfied with measurement, control and the requirements of electromagnetic compatibility testing of electronic equipment for laboratory No.26 part: the special requirements IVD medical devices measurement, control and the requirements of electromagnetic compatibility testing of electronic equipment for laboratory No.1 part: normal requirements standard requirements.
The following use requirements during operating should be obeyed, or it may cause electromagnetic interference by other devices or decrease the resistance to electromagnetic interference of this instrument, even lose the basic function.
According to the design and testing for group I list for this instrument, on the family environment, this device may cause radio

interference, which need adopt the prevention measurement.
The description for medical electrical equipment may affected by portable and mobile radio frequency communication equipment: the portable and mobile radio frequency communication equipment my affect the operation of this instrument, which guarantee the portable and mobile radio frequency communication equipment should have some space distance with this instrument, more specific requirements should follow the Table 4. Warning 1: In addition to the equipment or system manufacturer as a spare part for internal components sold outside the transducer and cable, use the component which outside the provision, transducer and cable may lead to the increase of equipment or system launch or ths decrease of immunity.
Warning 2: Equipment or system should not close or fold to use with other equipment, if it must close or fold to use, and then observe and verify the normal operation under this configuration.
Warning 3: Do not use this instrument beside the strong radiation source, or it mey interference the normal operation for this instrument.

Table 1

Induction and the declaration for manufactors--------Electromagnetic emission		
This instrument should use in the following specified electromagnetic environment, purchasers or users should use this instrument on this kind of electromagnetic environment.		
Emission experiment	Conformity	Electromagnetic environment--------induction
Radio frequency launch	Group 1	The automatic hematology analyzer only can use the RF energy for its internal functions.
Radio frequency launch	List A	The automatic hematology analyzer could use for all the facility which not for family and not directly connected to the home for residential public low-voltage power supply network
Harmonic emission	Not applicable	
Voltage fluctuation/ Flashing	Not applicable	

Table 2

Induction and the statement for manufactors---------Electromagnetic immunity			
This instrument should use in the following specified electromagnetic environment, purchasers or users should use this instrument on this kind of electromagnetic environment.			
Immunity experiment	Test level induction	Achieve the electrical level	Electromagnetic environment------- Induction
Electrostatic discharge	±4kV Contact discharge ±8kV Air discharge	±4kV Contact discharge ±8kV Air discharge	The ground should be woody, concrete or tile, if the ground covered by synthetic materials, the relative humidty should be at least 30%.
Electrical fast transient	± 1 kV pairs of power line	± 1 kV pairs of power line	Network power supply should have the quality of typical commerce or hospital environment
Surge	± 1 kV line to line ±2 kV line to ground	± 1 kV line to line±2 kV line to ground	Network power supply should have the quality of typical commerce or hospital environment.
Power input line voltage drops, immediately interrupt and voltage change	0% UT, continually 1 cycle (100% drops at UT); 40% UT, continually 5 cycles (60% drops at UT); 70% UT, continually 25 cycles (30% drops at UT); 5% UT, continually 5 seconds (30% drops at UT).	0% UT, continually 1 cycle; 40% UT, continually 5 cycles; 70% UT, continually 25 cycles; 5% UT, continually 5 seconds.	Network power supply should have the quality of typical commerce or hospital environment If the users of automatic hematology analyzer need continually operate it during interrupt period, and then recommend adopt UPS or battery supply.
Power frequency magnetic field (50/60HZ)	3A/m	3A/m	If the work is abnormal, it is necessary to keep the automatic hematology analyzer away from power frequency magnetic field or install magnetic shielding. It should measure the power frequency magnetic field of anticipate installation site and meet the requirements of less than coincidentdet level.
Attention: UT is the ac grid voltage before apply voltage.			

Table 3

Induction and the declaration of manufacturers-------Electromagnetic immunity			
The automatic hematology analyzer anticipate be used in the following electromagnetic, purchasers and users should guarantee use this analyzer in this kind of electromagnetic environment.			
Immunity test	Test electrical level	Achieve the electrical level	Electromagnetic environment------ Induction
Radio-frequency conduction Radiofrequency radiation	3V (Effective value) 150 kHz ~ 80 MHz 3 V/m 80 MHz ~ 2.0 GHz	3V(Effective value) 3 V/m	Portable and mobile radio frequency communication equipment should not closer than recommended isolation distance to the any parts of automatic hematology analyzer, which include cable, the calculation of distance should use the corresponding fomula for transmitter frequency. Recommended isolation distance d=1.2Vp d=1.2Vp 80 MHz ∽800 MHz d=2.3VP 800 MHz ∽2.0 GHz In the formula, P-----The maximum rated output power of the transmitter provided by the transmitter manufacturer, which use W as unit; d ---- Recommended isolation distance, which use m as unit. Field for fixed radio frequency transmitter measured by electromagnetic place a, which each frequency range b should less than achieve the electrical level. There may be interference near the device where has already marked the following symbol.
Attention 1: the formula for higher frequency should be adopted between 80 MHz and 800MHz. Attention 2: These inductions may be not suitable for all situations. And the electromagnetic transmission may effected by construction, substance and the absorption and reflection of human body.			
a. Fixed transmitters, such as, radio telephone (honeycomb/wireless) and ground mobile radio base station, amateur radio, AM/FM radio and TV broadcast etc, the field cannot be accurately predicted by theory. In order to assessing the electromagnetic environment of fixed radio frequency transmitter, which should considered the assessment for electromagnetic site. If the field measured by automatic hematology analyzer higher than the following RF compliance level, which should measure the automatic heatology analyzer to verify its normal working. If it detect the unormal function, so it is necessary to supplement measure, such as readjust the direction or site for automatic hematology analyzer. b. The field should less than 3 V/m between 150 kHz and 80MHz.			

Table 4

Recommended isolation between portable and mobile radio frequency communication equipment and automatic hematology analyzer			
The automatic hematology analyzer anticipate use in radioactive radiation harassment controlled electromagnetic environment. According to the maximum power output for communication equipment, the purchasers or users for automatic hematology analyzer could use the shortest distance between maintain portable and mobile radio frequency communication equipment (transmitter) and automatic hematology analyzer to prevent from electromagnetic interference.			
The maximum power output for transmitter (W)	The isolation distance for the corresponding transmitters/m		
	150 kHz-80 MHz d=1.2VP	80 MHz ∽800 MHz d=1.27P	800 MHz∽2.0 GHz d=2.3VP
0.01	0.12	0.12	0.23
0.1	0.38	0.38	0.73
1	1.2	1.2	2.3
10	3.8	3.8	7.3
100	12	12	23
For the maximum power output for transmitter which unlist in the above table, recommend isolation distance as d and unit as m, which could use the formula in the corresponding transmitter frequency list, P is the maximum power output of transmitter which supplied by transmitter manufacturer, which use W as unit. Attention 1: the formula for higher frequency should be adopted between 80 MHz and 800MHz. Attention 2: These inductions may be not suitable for all situations. And the electromagnetic transmission may effected by construction, substance and the absorption and reflection of human body.			

4.3 Analyzer Connection

Agappe authorized personnel should complete the following connections when installing the analyzer.

4.3.1 Pipeline Connection

- The handling of reagents, waste fluid, and waste samples, etc. should comply with the local and national regulations about the discharging and handling of bio-infectious materials like reagents, waste fluid, etc.
- Reagents may irritate eyes, skin and mucous membranes. The operator should comply with laboratory safe operation regulations and wear personal protective equipment (such as laboratory protective clothing, gloves, etc.) when using or touching related items of reagents.
- Once reagent touches the skin, rinse with plenty of water immediately and seek medical treatment if necessary.
- Once reagent touches the eyes, rinse with plenty of water and seek medical treatment immediately.

Warning

Important

- Do not place the reagents or other liquids on top of the analyzer.

Analyer tubing connections is shown as figure 4.3.1-1.

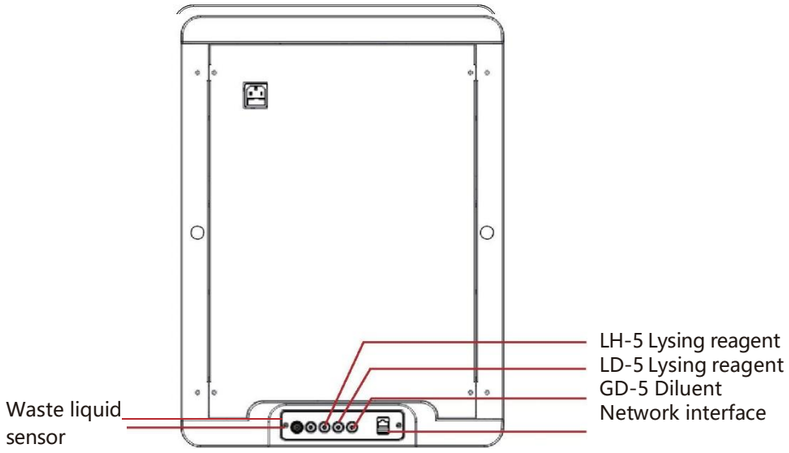


Figure 4.3.1-1

4.3.2 Optional Equipment Connection

The analyzer and optional equipment should be installed according to the instructions.
Make sure the connections are solid and reliable.

4.4 Software operating environment

- Hardware Configuration： CPU
  Intel Celeron J3060 1.6G 2C
  Memory: 4GB DDR3 1600 SoDIMM
- Peripherals: printer (optional), mouse, keyboard, Hand-held bar code scanner software configuration： 32 bit Windows operating system (Window 7 or 10),
- Net Framework 4.0 environment, MySQL Server 5.7 database, Office SoftwareNetwork： B/S structure,LAN,bandwidth:100Mbps

Chapter 5
Basic Settings

5.1 Overview

Analyzer has been initialized at the factory settings before leaving the factory, so parameters and reference values are at default.
Users can reset some parameters depending on the actual need. Setting Item

5.2 Setting Item

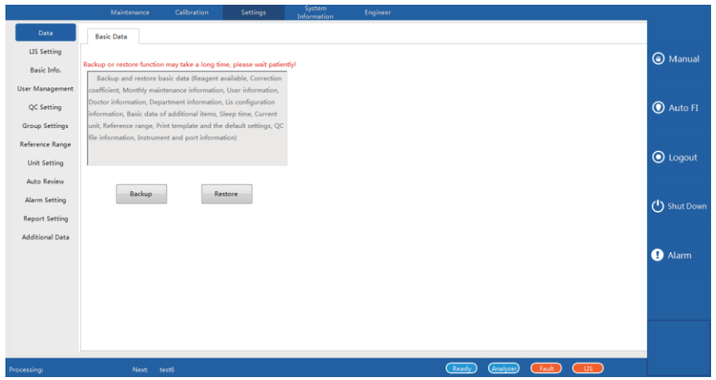


Figure 5.2-1

5.2.1 Report settings

Click "system" →"setting" — "report setting", and then enter into report setting interface, please follow as picture 5.2.1-1.

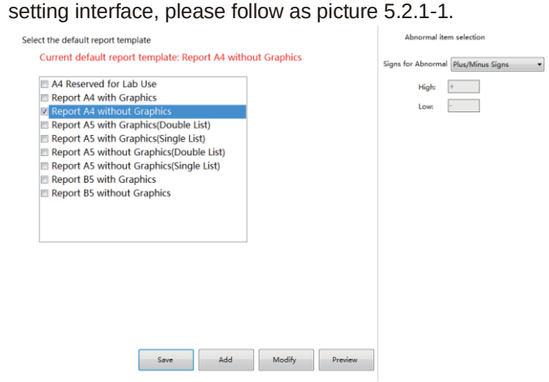


Figure 5.2.1-1

- Default Template Setting

It can choose software which already has worksheet template, select it and click “save”

- Report settings

→Abnormal Results Prompt Choosing "↑"" ↓" and "＋〞〞－〞 can be selected to prompt that the result
exceeds the reference range.

5.2.2 Reference Range Setting

After user starts the program, click System" - "Setting" - Reference Range" to set the reference ranges, such as Figure 5.2.2-1.

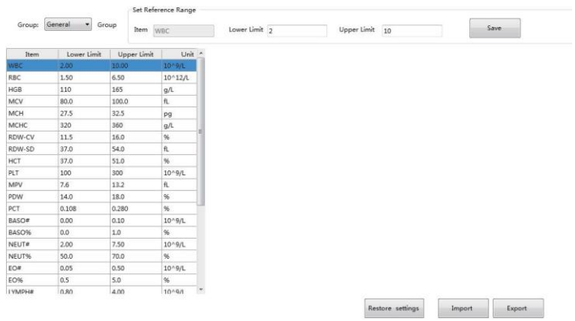


Figure 5.2.2-1

Click the item name, the item will be indicated by highlight, the upper and lower limit of current setting reference value will be indicated on the right side, modify the numbers in "upper limit" and "lower limit" input box separately, click and move to next item after finishing modification. And then click "ok" button to reserve after finishing all the settings.

- Group

  Click ,group" box, which could modify the different reference value group, system will supply 5 default groups and 5 custom groups, and also you could click "newly increase" button to add multiple custom groups.

- Restore to Default

  Click "Restore" and the system will restore the references value to factory setting.

  Restore Settings

- Click the "Restore Settings" button, the system will save the current reference value to the factory settings.

- Input

  Click "Input" and the reference values file selection window pops out. Select the existing ".xml" file format as reference values file. Such as Figure 5.2.2-2.

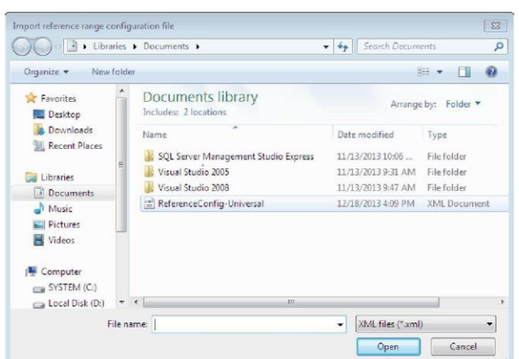


Figure 5.2.2-2

- Output

  Click "Output" and the saving interface pops out. It is shown in Figure 5.2.2-3.

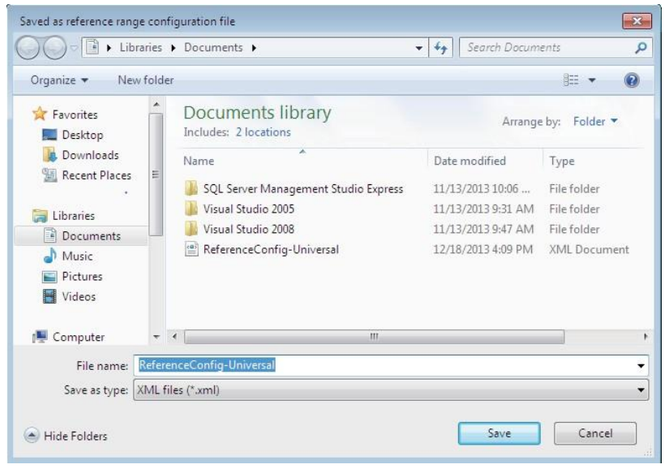


Figure 5.2.2-3

- Save

  Click "Save" to save the current modifications and exit reference value interface.

5.2.3 Reference Group Setting

After user start the procedures, click "system"- "settings" — "group settings" to proceed the classification settings of reference groups, please follow as picture 5.2.3-1.

Group Name	Age Lower Limit	Unit	Age Upper Limit	Unit	Gender	Effective
General		Blank		Blank	Unlimited	☑
Adult Male	14	Year	999	Year	Male	☑
Adult Female	14	Year	999	Year	F	☑
Children	29	Day	13	Year	Unlimited	☑
Newborn	0	Day	28	Day	Unlimited	☑
User-Defined1	1	Year	1	Year	Unlimited	☐
User-Defined2	1	Day	12	Day	Unlimited	☐
User-Defined3	1	Day	1	Day	Unlimited	☐
User-Defined4	1	Day	1	Day	Unlimited	☐
User-Defined5	1	Day	1	Day	Unlimited	☐

Figure 5.2.3-1

Important

- The default groups cannot be modified.
- When default groups are the same with the user-defined groups. User-defined reference value enjoys priority.

- Enable User-defined Group

  Click the "Effective" radio button, and select to enable the user-defined group.

- User-defined Group Modification

  Modify user-defined group age, unit, and sex.

Double click age to modify, which could modify the age number;
Double click age unit to choose the age unit of the group, with three kinds of "Year", "Month, "Day" and "Hour" available;
Double click "Sex" to choose the sex of the group, with three kinds of "Unlimited", "Male" and "Female" available.

524 IP Alarm Condition Setting Interface

Click "system" — "settings" — "IP alarm settings", which could modify IP alarm threshold, please follow as picture 5.2.4-1.

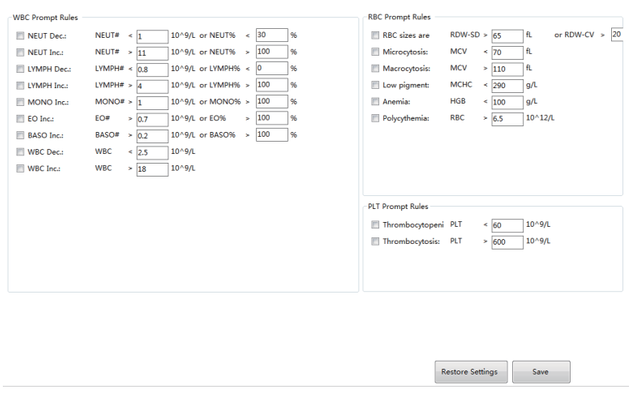


Figure 5.2.4-1

- Save
  Click "save", save current modification and exit current interface.
- Restore to Factory Setting
  Click "Restore" to restore the IP alarm threshold value to factory default setting.

5.2.5 User Management

Click "system" — "settings" — "user management", which could enter into user management interface, please follow as picture 5.2.5-1.

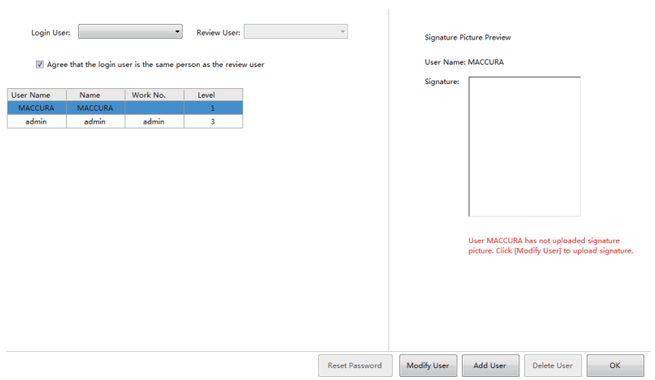


Figure 5.2.5-1

- Auto Login
  The account for automatically log in could be chosen in the automatically log in interface.
  Current account" indicate the account which have already set auto login. "Set account" could select the account which need auto login.

Important

- Low-privileged account cannot set high-privileged as automatic login account.
- When "User Setting" is disabled, it means that there is no automatic login account and the account name and password must be entered manually when starting the software.

- Reset password

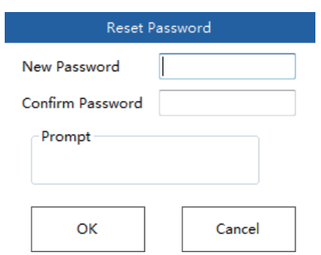
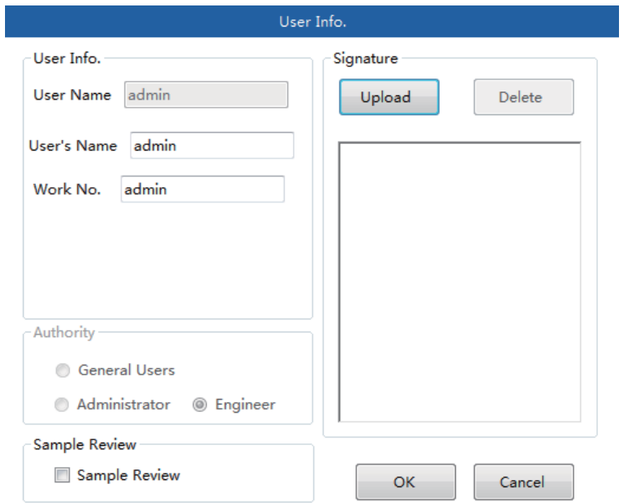


Figure 5.2.5-2

Select the user and click "Reset" to enter the new password twice. Click "OK" to save the new password.

- Modify User



■ Add User

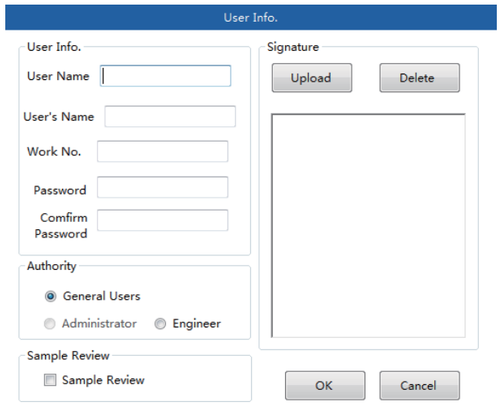


Figure 5.2.5-4

Click "Add User" to enter user creation interface.

Important

■ The first installed user account should be set up by engineer, and the general user can be set up by an administrator account or an engineer account

"User Name" is the account name entered when logging into the software; "Authority" is the new user' s level;

"Sample Review" is used to determine whether the new user has a sample review authority. If it is selected then the user has review authority.

■ Delete User

Click "Delete", and the following box will pop out.
Click "OK" to delete the user and exit the interface;
Click "Cancel" to exit this interface without deleting the user.

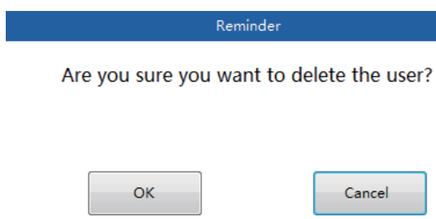


Figure 5.2.5-5

■ OK

Click "OK" to save the current modification and exit the delete user interface.

■ Cancel

Click "Cancel" will not save the current modification and exit the delete user interface.

5.2.6 Operation Log Management

Click "system" — "maintenance" — "Log" to enter operation log management interface, please follow as picture 5.2.6-1.

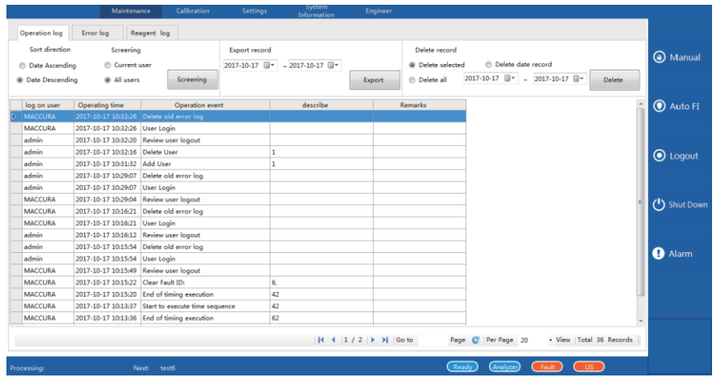


Figure 5.2.6-1

■ Filter

—Operation Date Ascending or Operation Date Descending can be selected in "Ranking Direction".

—Current User or All users can be selected in "Filter Requirements".

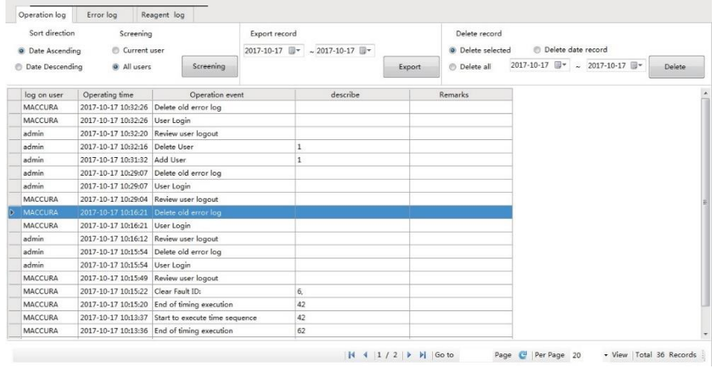


Figure 5.2.6-2

■ Export

Export Records can select the time period that needs exporting.

Click "Export" and the file saving window will be pop out.



Figure 5.2.6-3

■ Delete

— "Delete Selected Record(s)" can precisely delete the selected record(s);

— "Delete Record(s) within a Specified Time Period" can delete record(s) within the specified time period in bulk;

— "Delete All " can delete all operation records;

— Click "delete" and the deletion confirmation prompt box will pop out, click OK to delete corresponding record(s).

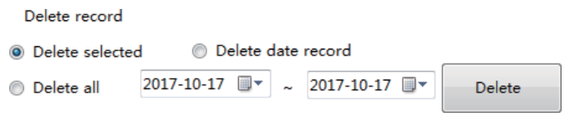


Figure 5.2.6-4

5.2.7 LIS settings

Click "system" — "settings" — "LIS settings" to enter LIS settings interface, please follow as picture 5.2.7-1.

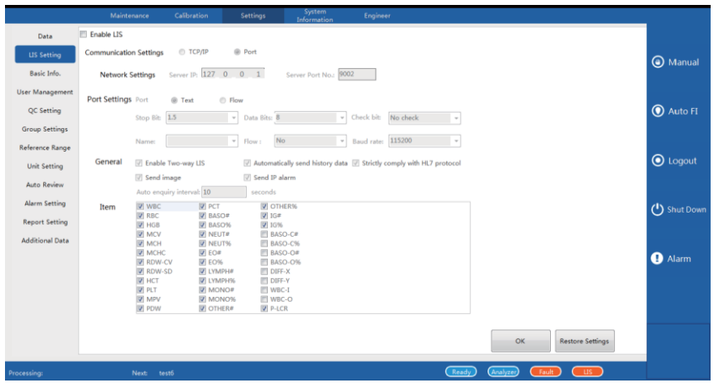


Figure 5.2.7-1

→LIS function

Server IP: input the IP address of the LIS server;

→Server port number: input the port number provided by LIS server;

→Enable LIS: selected means enabling LIS function.

→ Enable two-way LIS: selected means to obtain test mode and patient information from the server;

→Automatically send history date :selected means to send unsent result information in the history data automatically;

Automatic query interval: after enabling two-way LIS, automatic query interval can be set;

→Send images: whether WBC, RBC3 PLT histogram, and DIFF scatter plot are included in the packet, selected by default;

Send IP alarm item: whether the IP alarm item is included in the packet, selected default.

— Strict compliance with HL7 agreement: LIS data packets in strict accordance with the HL7 header format, the default is not selected.

5.2.8 QC

Click "system" — "settings" — "QC" to enter the QC setting interface, please follow as picture 5.2.8-1.

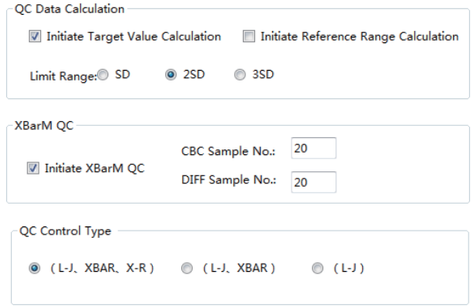


Figure 5.2.8-1

■ QC data calculation

There are two calculation methods to be chosen, such as "start the target value calculation" and "start the reference value calculation" , and it can also select all or unselect all.

Limit range can be selected from 2SD, 3SD and SD, only a single choice.

■ X-M QC

Select it and then start the X-M QC.

■ Ok

Click "ok" , and then reserve the current modification.

5.2.9 Additional data item

Click "system" — "settings" — "additional data item" to enter into additional data item settings interface, please follow as picture 5.2.9-1.

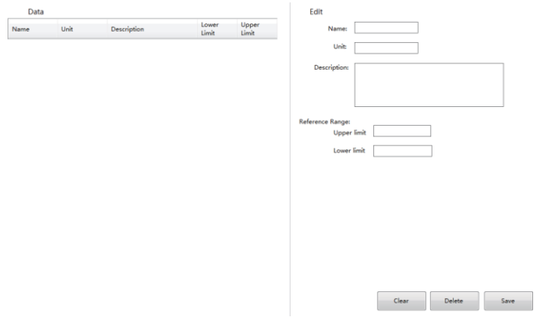


Figure 5.2.9-1

- Add data items
  Randomly select an original data item, modify the name, units, description and reference ranges, and then click "save" button.
- Delete data item
  Select the data item which need to be deleted, and then click “delete” button.
- Data item modification
  Select the data item which need to be modified, and then modify it.

Important

- The name cannot be changed during data item modification , or it will be regarded as newly increase date item.

5.2.10 Unit settings

Click “system" —"settings” — "unit setting" , the unit setting will be popped up, please follow as picture 5.2.10-1.

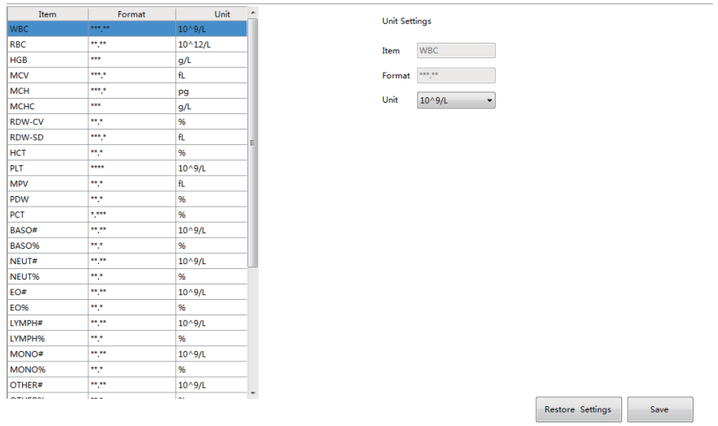


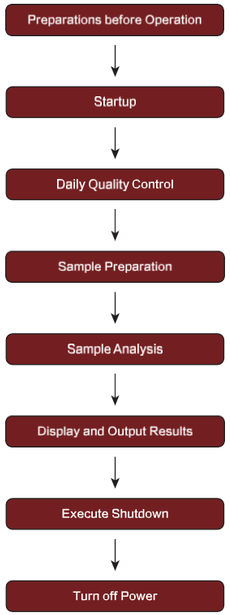
Figure 5.2.10-1

- Save
  Click “save" button, which could save the modification for units.
- Restore factory settings
  Click “Restore” button, and the parameter of current interface will reset to the initial value.

Chapter 6
Daily Operation

6.1 Overview

System contains a mainframe for testing, a set of supporting Mispa HX50 software for measurement, a computer for data processing, and a printer (optional).
This chapter introduces the daily operation of the instrument from startup to shut down.
This instrument only support run in manual sampling mode, which support only 4 measure modes, they are CBC+DIFF whole blood, CBC whole blood, CBC+DIFF pre-dilution and CBC pre-dilution.
The daily operation flow is as follows:



Preparations before Operation

Check the following to ensure that the system can operate normally before turning on the device.



- Samples, calibrators, control, waste liquid, etc. are potentially biological infectious. The operator should comply with laboratory safe operation requirements and wear personal protective equipment (such as laboratory protective clothing, gloves, etc.).
- This instrument within the liquid pipeline, it is recommended to check for leakage during each self-test run if so, immediately shut down and contact post-sale department of our company.
- In order to making sure the instrument is in safety status before use, please confirm whether the power supply, grounding conditions, waste discharge piping, etc. meet the requirements of this user manual or not.



- The handling of reagent, waste fluid, and waste samples, etc. should comply with the local and national regulations on bio-infectious materials like reagents and waste fluid, etc.

Do the following check before turning on the instrument:

1. Check the waste barrel
   Make sure that the waste bucket is empty before the test each day.
2. Check the reagents and connecting pipeline

- There are some differences between reagent amounts and analysis modes. Make sure that there is adequate amount of reagent for usage before startup every day, and prepare additional reagents for replacement. During the analysis, the instrument will stop automatically if the reagent is used up. If it happens, replace the used reagent.
  After the reagent has been replaced, start the analysis again.
- Make sure that the pipelines connected with the reagent bottle are not folded or twisted, etc.
  Make sure the power cord is in contact with the socket.
  Reagent consumption is as Table 6 - 1 shows.

Table 6-1 Reagent Consumption

Reagent Name	Single Sample Test (or other operate) Consumption	
	CBC+DIFF	CBC
GD-5 Diluent	≤ 32mL	≤ 26mL
LH-5 Lysing Reagent	≤ 0.38mL	≤ 0.38mL
LD-5 Lysing Reagent	≤ 1.2mL	0
DD-5 Fluorescent Dye	≤ 30uL	0
CC-5 cleaning solution	≤ 2mL/ Maintenance	

3. Check printer (Optional)
   Check whether printer paper is enough and whether power connection with computer is well.
4. Check the network communication connection for instrument and computer is normal or not.

6.2 Startup

Start the computer and open the Mispa HX50 software, switch to "on" position on the right of the Analyzer mainframe. After power on, the indicator light will on light.

Important

- You can either start Mispa HX50 software first or the Analyzer mainframe first. There is no required order.

When the mainframe and the Mispa HX50 software have been started, both of them will connect automatically. The system will perform startup sequence after successful connection, including version checking, device cleaning, time sequencing loading and background checking.
When the time sequence of the device cleaning is completed successfully, the instrument will automatically enter the background checking, which is the measurement of particle interference and electromagnetic interference.
If the first background results obtained during the initialization of the liquid path exceeds the background range, the analyzer will performs another background detection automatically. The results of background detection will not process high-low or suspicious alarm.
After finishing background detection, and then process temperature detection.
If startup fails, the buzzer will give off alarm and the software displays the warning dialog box. User can operate according to the instructions of the dialog box. If the problem is still unresolved, please contact to the customer service department of our company.

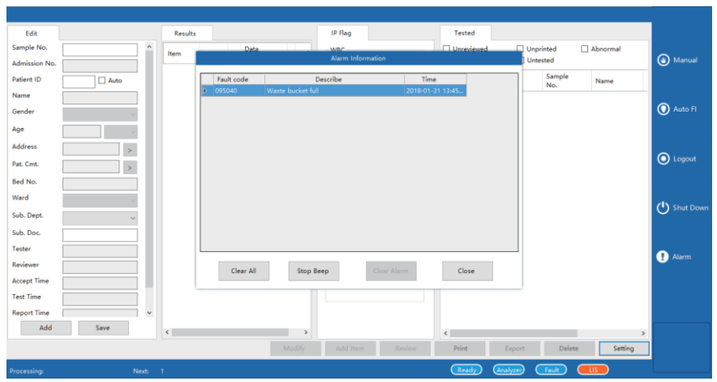


Figure 6.2-1

After finishing the initialization procedure, the system get into the main interface, such as following figure 6.2-2.

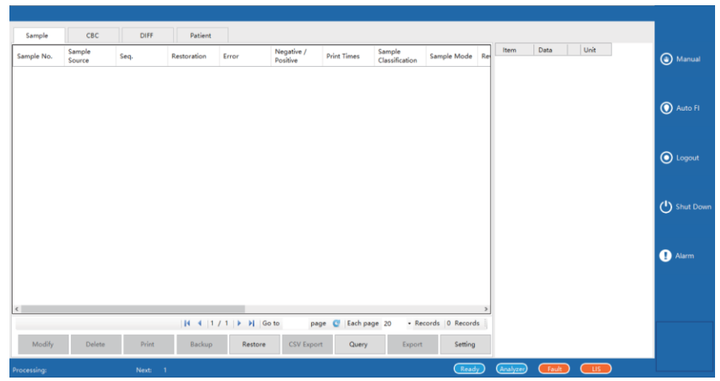


Figure 6.2-2

6.3 Quality control

Before sample analysis, the device requires daily QC to ensure reliable results. See the corresponding chapter for detailed quality control methods.

6.4 Analysis mode

This instrument only support the following analysis mode:
Manual sampling mode: on this operating mode, user mix the samples manually and then put them in manual sample position, instrument will perform the test.

6.5 Sample Preparation

- Samples, calibrators, control and waste liquid, etc. is potentially biological infectious. The operator should comply with laboratory safe operation requirements and wear personal protective equipment (such as laboratory protective clothing, gloves, etc.).

Important

- Do not contact patient's blood sample directly.
- All performance results in this user manual were obtained using EDTA anticoagulants in whole blood samples. The use of other anticoagulants will affect the results. Please ensure

6.6 Work List Information Registration

If the match method for sample is "non-scan generation" match method, worksheet sample need to be registered before measure them. Open software
In "work station" interface, please follow as picture 6.6-1.

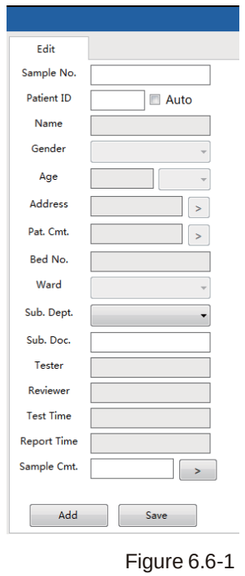


Figure 6.6-1

Click "add" button, edit sample information. Sample No. (After clicking "add", it will be automatically generated and can be modified). The measurement mode is required input (click the add button, the information will be automatically generated, the user can also modify). The information of the patients, the departments and the inspection doctors can be and the inspection doctors are all optional (the information can be modified before or after the measurement).
After the sample is edited, click the "save" button. After successful saving, it will be displayed on the "sample to be tested[11] sheet on the right.

Important

- Manual sampling match with sample number, for example, when the current measuring worksheet sample number is W1 the next automatically matched worksheet sample number is W2; when the test worksheet do not have "W2", match is failed, user need manually set the next worksheet sample or use the automatically generated sample after finishing measure procedure.
- "BLANKCHECK" "CORRECTION"" and the sample number start from "QC-is corresponding background
  test sample, calibration sample and QC sample separately. Therefore, register worksheet sample cannot use these
  three special sample number.

6.7 Whole Blood Samples

The preparation method of whole blood samples:

1. EDTA-K_2 EDTA-K_2 EDTA-Na (1.5 ∽2.2mg/mL blood) Anticoagulation vacuum blood collection tube is used to collect vein blood samples.
2. After collection, immediately shake the collection tube upside down several times to mix the blood with anticoagulant thoroughly.

Important

- In manual whole blood mode, blood sample must be no less than 1mL.
- The samples needed for WBC classification and PLT count should be stored at ambient temperature and analyzed within 4 hours after collection. If not, please store it within 2℃ ~ 8℃ freezer, until can begin analysis.
- If the results of WBC classification, PLT count and MCV are not needed, the samples can be stored in the refrigerator with
  the temperature of 2℃ ~ 8℃ for 24 hours.
- The frozen sample should be placed in the ambient temperature for 15 minutes to 30 minutes before analysis, and then lightly shake samples 10 times.
- Capillary blood is collected into a microcatheter to which anticoagulant was added.
- After standing for a while, the sample needs to be re-mixed before be analysis.

6.8 Pre-diluted sample

Pre-diluted sample preparation methods:

1. Click auto Fl button in software, please follow as picture 6.8-1.



Figure 6.8-1

2. It will pop up the automatic injection dialog box, please follow as picture 6.8-2.

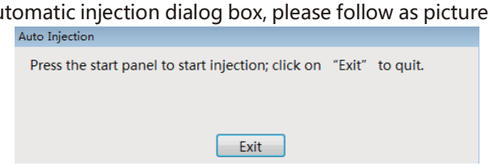


Figure 6.8-2

3. After finishing qutomatic sampling procedure, and then the instrument will automatically perform the automatic injection procedure, please follow as picture 6.8-3.

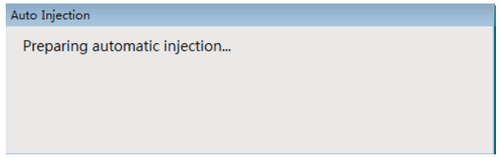


Figure 6.8-3

4. Put the tube in the manual sampling position, press "start" button, and then the instrument will perform automatic sampling procedure, such as figure 6.8-4.



figure 6.8-4

5. After the injection is completed, click the "exit" button to exit the auto injection, and the prompt window will pop up, as shown in figure 6.8-5.

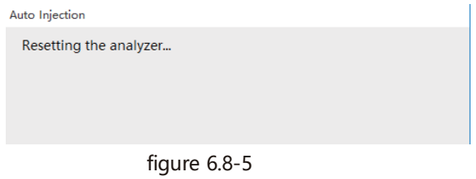


figure 6.8-5

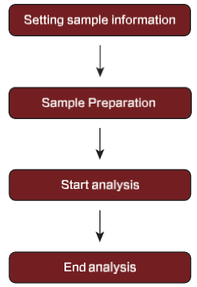
6. Manually mix the 20 u L peripheral blood and 200 R L diluent outside the instrument.
7. After the sample is mixed thoroughly, open the lid and place the mixed samples under the aspiration pipette carefully. Press the start button, and the pipette will draw about 91.5ul diluted sample for analyzing, and display the results of the analysis which has been multiplied corresponding times.

Important

- The mixture of capillary blood and dilution should be placed minutes before the test.
- Make sure that the capillary blood samples are tested within 30 minutes after dilution.
- Laboratory should evaluate the stability of the test results under pre-dilution mode according to its the number of sample, sample collection methods and technology.

6.9 Manual sampling analysis

Operation workflow as follows:



Specific operation steps are as follows:

1. After well prepared, sample information need to be set by "manual Sampling" button on the right of software menu area (please follow as picture 6.9-1).

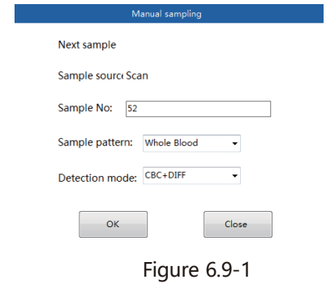


Figure 6.9-1

2. Shake the tube firstly, after mixing the blood totally, place the capped blood sample under the manual injection device.
3. Click "manual sampling" button on the instrument to start measurement.
4. After finishing suck blood, please take out the sample from the manual injection device.

- After the sample measurement begins, the matching method of the sample is based on the first sample. If work sheet match mode, automatically according to the order of matching a sample (such as single sample of 1 and 2,1 after the start of the 2-automatic measurement, as the default sample, if not registered *2,* matching failure); if the automatic matching method, with +1 type production a sample (such as the current measurement of the sample
number is S1 is a default manual sampling sample number for the S2. In the way of scanning matching, the sample number is not generated. It is necessary for users to manually input barcode (barcode number as sample number). No barcode is allowed to be measured.

Important

- Manual sample measurement supports continual measurement, after finishing the blood absorption of the previous sample, take out the sample, and then put the second sample into manual sampling device and press the measurement button. When continually measuring, the measurement mode of the samples must be consistent.
- When it registers the worksheet sample of manual sample, sample No. should be continuous, which could be easy to automatically match next worksheet sample during measurement procedure, which could reduce manual setting and improve work efficiency.

6.10 Results View

After finishing sample analysis, the results could be seen in "tested sample" list in the working interface, please follow as picture 6.10-1.

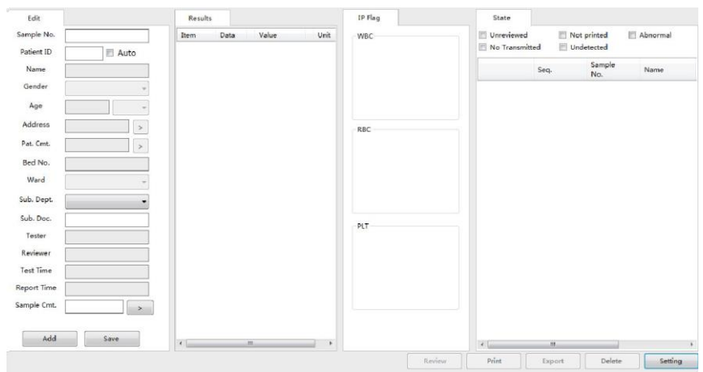


Figure 6.10-1

After clicking the sample, the analysis results are displayed in the middle area. If we need to modify the information of the tested samples and edit directly in the left edit window, we must input the patient's ID information when we need to input the patient information.
Sample analysis results could also be checked in history data and detailed browse function, more specific information could be seen in results examine chapter.

6.11 Shutdown and exit procedure

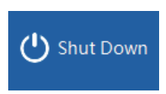
If the device is in continuous operation, shutdown operation must be performed at least every day after the end of the analysis or every 24hrs. The tubes, detection and analysis components of the device will be cleaned when the device performs shutdown operation.

Important

- To ensure reliable and stable performance of the analyzer, please make sure the following steps are followed to perform shutdown process.

Shutdown operation steps:

1. When the analyzer is ready, click the Shut Down button in the task bar of the system function.



2. The screen pops up the prompt window as picture 6.11-1 .



Figure 6.11-1

3. Click "shut down" button, the instrument will perform shutdown procedure. After shutdown process is completed, Power off interface will pop out as picture 6.11-2shows.

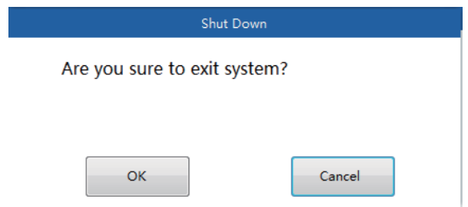


Figure 6.11-2

4. Click "OK" button, the analyzer will perform the shutdown process. After the shutdown process is completed, the system prompt will pop out, as shown in

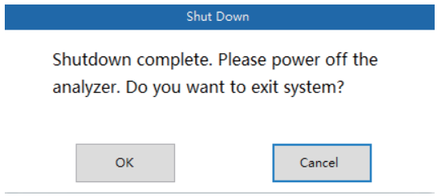


Figure 6.11-3

5. Click "OK" to exit the system software;
Click "Cancel", the system software will continue to operate offline.
6. Turn off the power switch on the right of the analyzer.
7. Empty the waste bucket, and handle the waste solution and waste samples properly.



- Samples, calibrators, control, and waste liquid etc. is potential biological infectious. The operator should comply wit laboratory safe operation requirements and wear person protective equipment (such as laboratory protective clothin gloves, etc.)

Important

- After the device is power off, user can still use the software functions of results checking, results printing and other non-interactive features with the host machine.
If it needs to operate software continually after shutdown procedure, next time you start the machine, be sure to restart the software.



Chapter 7 work station

7.1 Overview

Click ” work station “button to open the work station interface for sample registration, result view, etc., as shown in figure 7.1-1.

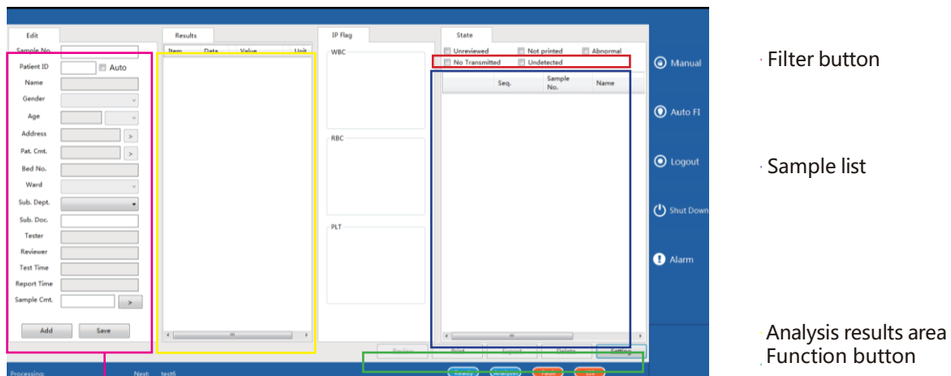


Figure 7.1-1

7.2 Function button

- “Audit”

Audit the selected tested sample.

- “Print and preview”

Select one audited sample, click "print” button to preview or print.

- “Export”

Select the results that need to upload to LIS, click "export" to upload the data to the LIS server.

- “Delete”

Select the sample which need to be deleted, click “delete" button to delete the sample.

Important

■ When editing the sample information, it is possible to make some items invisible if they are not needed (eg. patient ID, patient address).

- “Setting”

It can set these items invisible, such as figure 7.2-1.

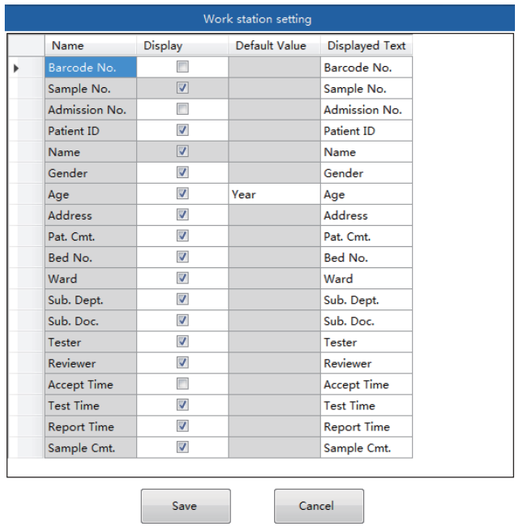


Figure 7.2-1

- Patient ID is unique and unrepeatable, system identifies patients by patient ID. When patient ID is selected as invisible$_3$ the system will automatically generate patient ID. When it is selected as visible, patient ID must be entered when input patient information.
- Only audited sample could be printed out, audited sample can cancel audit, printed sample can cancel audited as well.

7.3 Sample editing area

1. Worksheet information input

- Click "work station" button in main menu, which is on the left side of the menu, as shown in picture 7.1-1. Click “add” button, input sample information, after that click "save" button, and then the current sample information can be saved.
- Sample number, sample mode and test mode are the necessary options, other items can be configured.
- When it needs continuously add worksheet sample, click “add" button continuously.

2. Sample modification

- Click and select one sample.
- Modify sample information in sample editing area directly.
- After finish modification, and then click “save" button.

Important

■ Only worksheet sample did not test and have already tested but sample did not audited can be modified.

7.4 Sample list

Work station sample list area shows sample to be test and 1000 tested regular samples in last 3 days. Quality control, background checking and imported samples are not displayed.

7.5 Analysis results area

Select one tested sample in sample list, it will display sample test results and IP flag information in analysis results area.

7.6 Filter button

After selecting tested sample in sample list, there will be a "filter" button

on the top. Filter conditions consists of unaudited, imprint, abnormal, not transferred and test sample status.

- When it starts the procedure, three buttons cannot use, button is in "selected" status after click the button, which means to use this filter condition. Click again to cancel this filter condition.
- These filter conditions can choose single or multiple conditions simultaneously, or do not choose them all, more specific filter based on the requirements of users.
- "Abnormal" refers to sample which alarmed by IP flag, as long as one Alarm item occurs it can be regarded as abnormal sample.



Chapter 8
Results view

8.1 Overview

The analysis results will be saved to the database automatically each time. System databases can save 100,000 results including information of 28 parameters, 3 histograms, 1 2D scattergram and results display.
Users can choose to view batch of analysis results in the manager mode or to view a detailed analysis by the way of browser (Display histogram and scattergram).

8.2 History data

Click "history data[55] button in main menu, shown as picture 8.2-1.

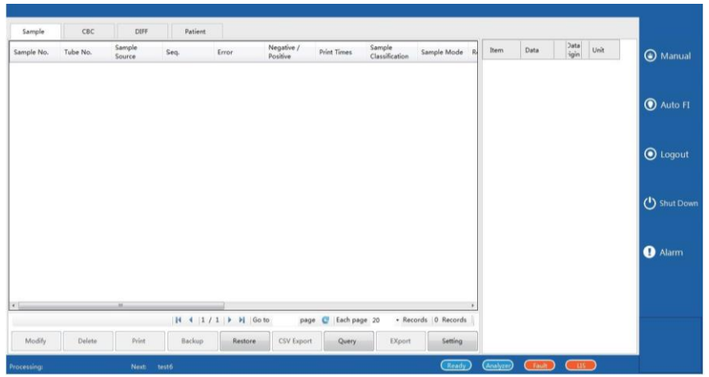


Figure 8.2-1

8.2.1 Results view

Users can view the results saved by the system in this interface. All the results will be arranged in chronological order, with the latest analysis results displayed at the top of the list.
If the storage of the results exceeds the display range of the screen, a scroll bar will appear at the right of the list, and the user can drag the scroll bar to turn the page and view the results.

8.2.2 Results Select

- Single Result Selection
  User can select the analysis results that need viewing or further operation (left-click or click arrow on the menu bar) based on his/her own needs. The selected results will be highlighted as Figure 8.2.2-1 shows.

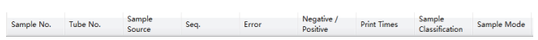


Figure 8.2.2-1

After the current result is selected, the analysis results will be displayed in the right results column as Figure 8.2.2-2 shows.

Item	Data		Unit
WBC	++++		10^9/L
RBC	10.35	@	10^12/L
HGB	0	-	g/L
MCV	----		fL
MCH	----		pg
MCHC	----		g/L
RDW-CV	----		%
RDW-SD	----		fL
HCT	++++		%
PLT	919	+	10^9/L
MPV	----		fL
PDW	----		%
PCT	----		%
BASO#	----		10^9/L
BASO%	----		%
NEUT#	----		10^9/L
NEUT%	----		%
EO#	----		10^9/L
EO%	----		%
LYMPH#	----		10^9/L
LYMPH%	----		%
MONO#	----		10^9/L
MONO%	----		%
IG#	----		10^9/L
IG%	----		%
WBC-I	++++		10^9/L
WBC-O	0.01	-	10^9/L
P-LCR	----		%

Figure 8.2.2-2

Users can also click "Sample Info ", " CBC", "DIFF" and "Patient Info." in the lower left corner of the manager, select the corresponding column table for viewing, as Figure 8.2.2-3 shows.

- Sample Info, column shows the sample number, date, time of the test, the test mode and other information;
  "CBC" column displays the CBC item results of this test item and results prompt;
- "DIFF" column displays the DIFF item results of this test item and results prompt;
- "Patient Info" column displays the name, sex and others information of patients.
- Some common columns (including sample number) often display in the left side so that users can conduct accurate search when switching columns.



Figure 8.2.2-3

- Multiple Results Selection
  Users can select the results of in bulk if needed. Press the left mouse button and drag in the management interface to select the results continuously. Press "ctrl" on the keyboard to select the needed results.

Important

- All batch results will be highlighted, and the result column will only display the sample pointed by the arrow at the most left side.

8.2.3 Audit

The sample results cannot be audited in the historical data interface.

Important

■ The quality control data which the sample number tarted With "QC- " and the quality control data which the sample number is "correction" detected by blank detect data which sample number is "BLANKCHECK" , which could not modified the operation. The quality control data which the sample number started with "QC- "and the blank detect data which the sample number is "BLANKCHECK" , which can modify the operation.

■ Reviewed status displays "S" mark in review column.

8.2.4 Results Modification

8.2.5 Results correction

Users are not allowed to perform a result correction for any sample in the history data interface.

8.2.6 Results Deletion

Users can delete unwanted data according to his/her needs. The operation procedures are as follows:

1. Select the single sample or multiple samples which need to be deleted.
2. Click "delete" button at the right comer of the bottom.
3. The dialog box will be popped up as picture 8.2.6-1.

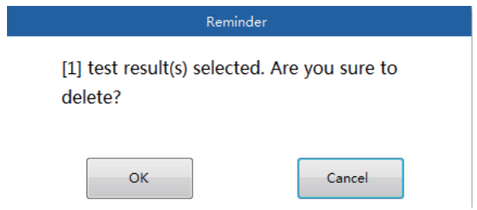


Figure 8.2.6-1

4. Click "ok" button to confirm delete, click "cancel" button to cancel the data deletion procedure.

Important

■ Audited sample cannot be deleted.

8.2.7 Upload the results

Select the LIS results which need to be uploaded, click "Export" and the data will be uploaded to the LIS server automatically.

Important

■ Sync feature can only be used when LIS is successfully connected.

■ After the test is completed, new sample will be uploaded to the server automatically.

8.2.8 Report Print

Users can print out reviewed results in the history data interface as follows:

1. Select the result needs printing. Click "Print" , as Figure 8.2.8-1 shows.
   The successfully connected printer will start printing the report list.

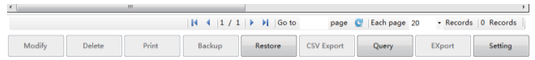


Figure 8.2.8-1

2. For a preview, click ,print" button, print preview interface will pop out as Figure 8.2 8-2 shows.

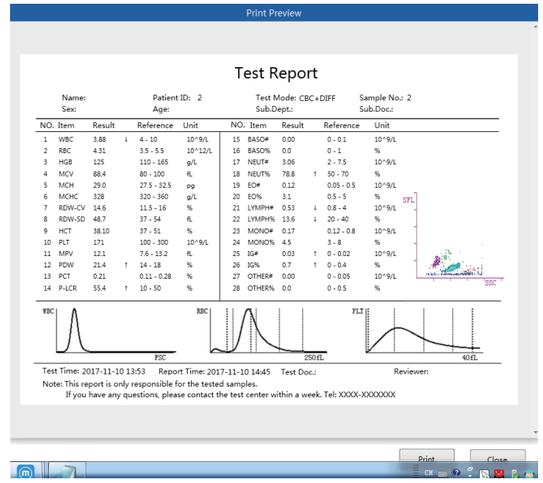


Figure 8.2.8-2

Important

■ Confirm printer installation, connection status, and whether paper supplies are sufficient before printing the report.

■ Not reviewed samples are not allowed print preview or print output.

8.2.9 Results back-up

User can back up results of selected historical data. Details for operation steps as follows:

1. Click "history data" — "back-up" button, shown as picture 8.2.9-1.



Figure 8.2.9-1

2. Choose the backup path for the selected documents in dialog box, shown as picture 8.2.9-2.

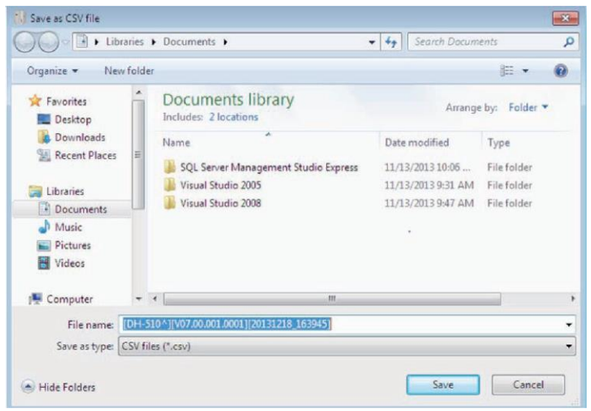


Figure 8.2.9-2

3. Click "Save" button to confirm backup, click "cancel" button to cancel current operation.
   Backup content consists of the sample information for the historical data, test results and graphics and patient information. User can select one data or multiple data to backup, and one data will generate only one SMP format document.

8.2.10 Results restore

User can recover the data which has already backed up, the details operation as follows:

1. Click "history data" — "restore" button, shown as picture 8.2.10-1.



Figure 8.2.10-1

2. Choose the path for the documents which need to be restored in dialog box, shown as picture 8.2.10-2, and then click "ok" button to restore.

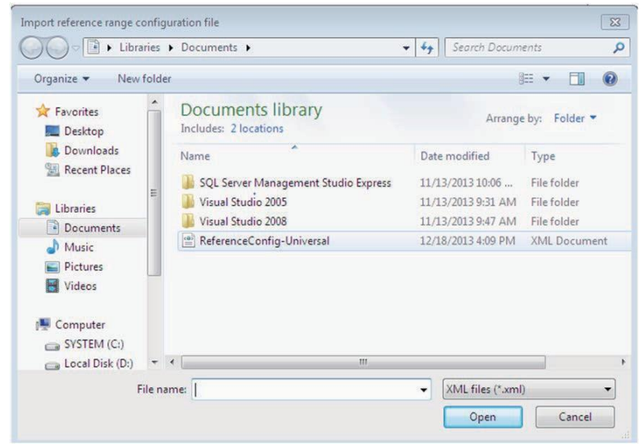


Figure 8.2.10-2

Important

- The sample restore function cannot be used while the instrument is running.
- The restored sample is labeled "Import." Imported samples cannot be audited and modified.

8.2.11 Results Export

Users can export single or batch results to a computer in CSV format in the manage interface, so as to use EXCEL and other statistical tools for sample analysis. Abnormal Results Prompt Choosing.

Export methods are as follows:

1. Click "history data" - "CSV output" button, shown as picture 8.2.11-1.



Figure 8.2.11-1

2. The dialog box for need export pictures or not will appear, shown as picture 8.2.11-2,select output pictures or not as needed.

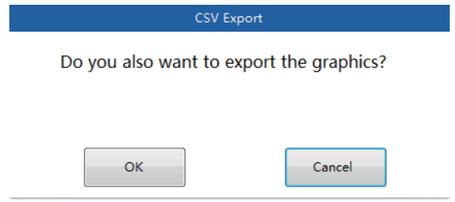


Figure 8.2.11-2

3. After click dialog box for need output pictures or not, the dialog box for output target location will appear, shown as picture 8.2.1 1-3.

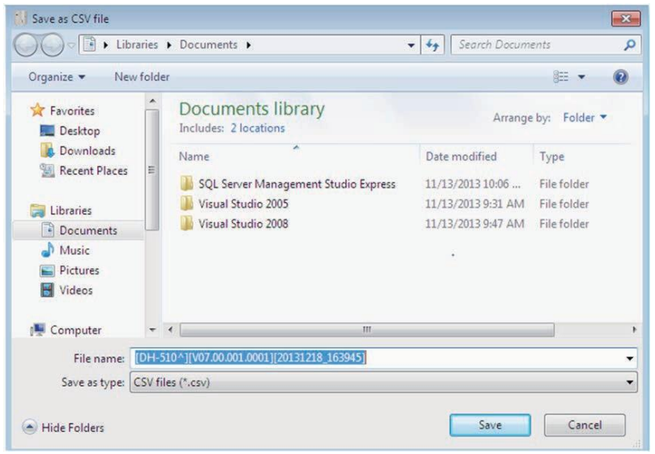


Figure 8.2.11-3

4. Click "save", the selected results will reserve in target location with the document name which named by system automatically in CSV format.
5. The dialog box will appear after finishing output, shown as picture 8.2.10-4.

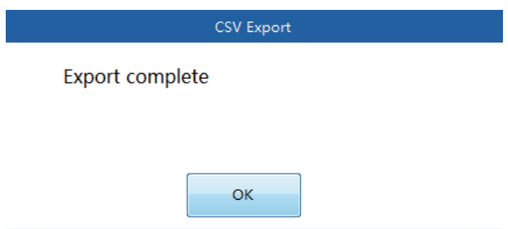


Figure 8.2.11-4

6. Click "ok" button, and log out this dialog box.

Important

- The document name automatically generated by the system, and user can rename the document as needed by themselves.
- Save type is CSV format only.

8.2.12 Result query

User can filter the qualified sample in history data. The details as follows:

1. Click "history data" - "query" button, shown as picture 8.2.12-1.

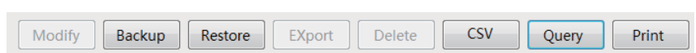


Figure 8.2.12-1

2. Conditional filter dialog box will appear after click "query" button, shown as picture 8.2.12-2, user can query data by selecting query condition.

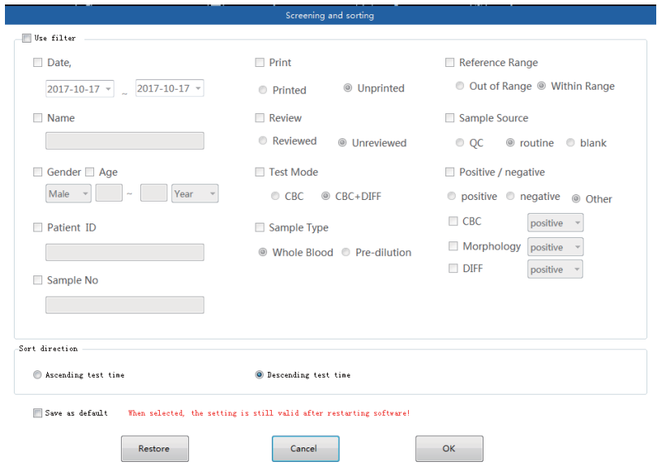


Figure 8.2.12-2

■ Query conditions

In picture 8.2.12-2 interface, select "use filter" button, and then further select the conditions which need to query simultaneously.

→"Date" can be selected by date method to query.

→ "Name" can input patient name to query.

→ "Patient ID" can input patient ID to query.

→ "Sample number" can input sample number to query.

→ "Print" can be selected by printed, not print two methods to query.

→ "Audit " can be selected by audited, not audit two methods to query.

→ "Test mode" can be selected by CBC+DIFF mode, CBC mode, predilution CBC+DIFF mode and precaution CBC mode.

■ "Reference range" can be selected by outside reference range and inside reference range to query.

→"Sample source "can be selected by quality control, routine sample and background checking to query.

→ "Negative/Positive" can be selected by negative sample mode and positive sample mode.

→ "Sort direction" can select detection date by descending order and

Important	■ Query function can be selected by choosing single condition and multiple conditions to query. ■ When selecting measure time, the input time cannot more than the current date, it prompts the user that the query is not executed.

8.3 Detailed browse

Click "detailed browse "button in main menu, then the detailed browse interface will appear, shown as picture 8.3-1.

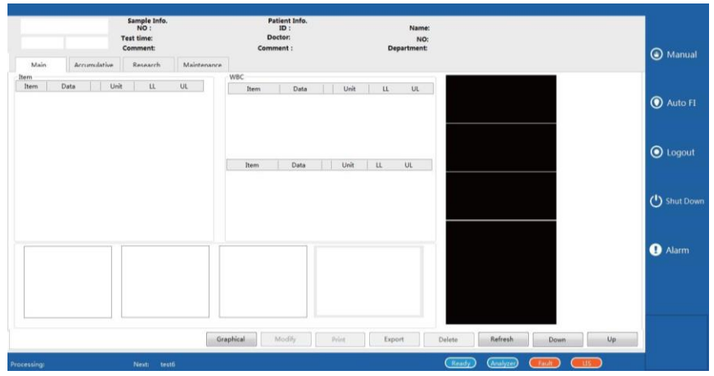


Figure 8.3-1

User can check the single sample result briefly in "detailed browse" function, which includes histogram, scattergram and negative/positive alarm information. User can select one tested sample in worksheet or history data interface, and then click "detailed browse" button in tool list to browse sample briefly. User also can click "detailed browse[15] button in main menu to check the latest tested sample directly.

8.3.1 Results Checking

The browser displays the currently selected sample analysis results, negative or positive alarm, audit status, sample information, patient information, Flags tips and other information.

Users can view the results by clicking "Upper" / "Lower" on the main menu, can also use the T 1 keys on the keyboard.

Click "Research" and the research interface will open as Figure 8.3.1-1 shows. In addition to the 24 report parameters displayed in the research interface, 6 research parameters are also displayed, such as IG #, IG%, OTHER#, OTHER%, WBC-O, WBC-I.

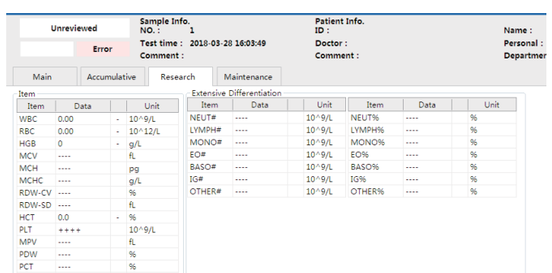


Figure 8.3.1-1

Important

- Double click the figure (3 histogram and 1 scatter plot) for magnification.

8.3.2 Related instructions for detailed browse interface

Samples which did not audited cannot be modified or audited in detailed browse interface, but they can be deleted; Reviewed sample can be selected to view, preview, output and print data in detailed browse interface, but they cannot be deleted.

Important

- Sample number starting with "QC-" is QC data and the sample number is "BLANKCHECK" and the sample number of black test data is "correction" , they are cannot be modified.
- Sample number starting with *"QC-"* is QC data and the sample number is "BLANKCHECK", which can be deleted.

Important

- Reviewed results will be displayed with a green background in the status bar, and the results which have not been reviewed will be displayed with a white background in the status bar.

8.3.3 Results lock

Click "detailed browse"— ''refresh" button, which could control the detailed browse interface will automatically refresh or not when the latest sample result released.

1. When the button selected as "locked" , shown as picture 8.3.3-1, the detailed browse interface will not automatically refresh when the latest sample result released.

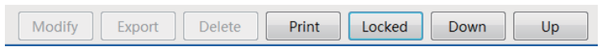


Figure 8.3.3-1

2. When the button selected as "refresh", shown as picture 8.3.3-2, the detailed browse interface will automatically refresh when the latest sample result released.



Figure 8.3.3-2

Chapter 9 Analyzer Calibration

9.1 Overview

The purpose of the analyzer calibration is to ensure accuracy of the analysis results.

Warning

- Please pay attention in operation since aspiration pipette is fine, and the calibrator, control and biological samples it carried may be biologically infectious.
- The handling of waste calibrator, package of QC material should comply with the local and national regulation on th discharging and handling of bio-infectious materials like reagent waste fluid, etc.

- Samples, calibrator, control, and sample is potentially biological infectious. The operator should comply with laboratory safe operation requirements and wear personal protective equipmen (such as laboratory protective clothing, gloves, etc.).

Important

- Only login users with administrator authority can conduct
- Users should use the quality control material and calibrators specified by Agappe, and store the calibrators, quality control materials and reagents in the strictly instructed storage environment.

After the calibration, the analyzer can ensure the accuracy of the analysis results for a long time, but under the following conditions the analyzer may need new calibration.

- Before using the analyzer which leave unused for a long time Analyzer has not been used for a long time, before using;
- After the replacement of the main components;
- Observe quality control data, when deviations occur on quality control data confirms after double check.

Analyzer provides manual calibration and automatic calibration.

Please confirm the following conditions before calibration:

- Analyzer should meet the requirements of the background as appendix shown. If the background appears abnormal, please refer to fault handling procedure for solution. If it cannot be resolved, please contact Agappe 's customer service;
- Remaining reagents are sufficient to complete the calibration. If reagent shortage occurs during calibration, it should be recalibrated;
- Analyzer repeatability should meet the requirements shown in appendix. If repeatability appears abnormal, please refer to fault handling procedure for solution. If it cannot be resolved, please contact Agappe' s customer service.

Important	■ Repeatability test method: under the whole blood test mode, use mid-value quality control blood to count 10times continuously, and then the results will be exported in CSV format and then use Excel to count CV value.

9.1.1 Manual calibration

Manual calibration is as follows:

1 Prepare calibrator or fresh blood that has been tested on the reference analyzer to obtain reference value.

2 Enter the sample test interface, select CBC + DIFF mode or pre-dilution CBC+DIFF mode.

3 Test calibrator or reference values obtained fresh blood in the analyzer 11 times continuously, and calculate average of 2nd to 11th and then get the measurement average value.

4 Double click system[55] — "calibration" and select manual calibration.

5 Manual calibration interface will pop out as Figure 9.1.1-1 shows.

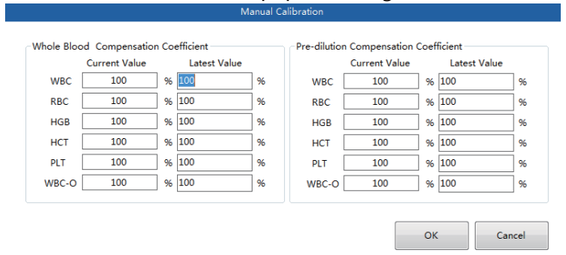


Figure 9.1.1-1

6.Enter the latest correction factor in the corresponding compensation t item.

7.Click" OK" to save the new modified coefficient automatically and exit the current interface.

Important

- Only login users with administrator privileges can calibrate.
- If fresh blood is not used, user should use quality control material and calibrators specified by Agappe and store the calibrators, control and reagents in the strictly instructed storage environment.
- The correction and calibration mentioned in this article are the same.

9.1.2 Auto Calibration

Automatic correction method is as follows:

1. Prepare calibrators or fresh blood that has been tested in the reference analyer to obtain the reference value.
2. Click "system" — "calibration"— ''automatic calibration", and then pops up the automatic calibration interface, as Figure 9.1.2-1 shows.

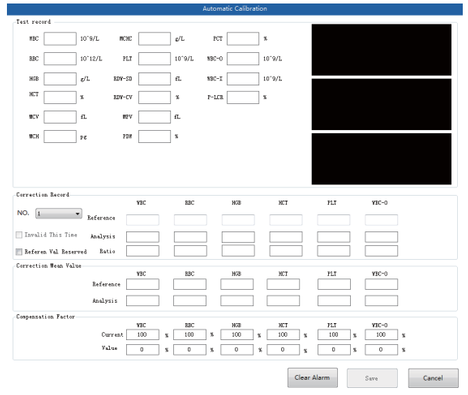


Figure 9.1.2-1

3. Test calibrators or reference values obtained fresh blood on the analyzer should be tested at least 5 times continuously, and each reference should be entered into the reference value input box. Select NO. values as "+1 " after each test. If you are using the same blood for calibration, you can choose "Retain Reference Value" to avoid repeated input of reference value.
4. If test times are more than or equal to 5 times, the software will calculate a new correction factor automatically, click "Save" to save the calibration results.
5. Apply the following formula to calculate the new calibration coefficient.

$$\text{The new calibration coefficient} = \frac{\text{current calibration coefficient x reference value}}{\text{measurement average}}$$

Important

- There will be a corresponding message if the results do not meet the specification. Please check referring to the

9.1.3 Correction Factor Confirmation

After system calibration, the new correction factor should be confirmed according to the following steps:

Step 1: Test the calibrators three times to make sure the analysis results are within the allowable range.

Step 2: Run the high, medium and low three concentrations of control, test each concentration three times to make sure the analysis results are within the allowable range.

Step 3: Test three normal fresh blood samples whose results are already known, each sample will be tested three times to make sure the analysis results are within the allowable range.

Chapter 10
Quality control

10.1 Overview

In order to monitor the performance of analyzer in operation, performing quality control regularly is required. The laboratory should develop standardized quality control operating procedures and ensure its strict execution.
After calibration or main components replacement of the analyzer, additional quality control is required.

Important

- Use specified control blood. Store the quality control material in the strictly instructed storage environment.



- Samples, calibrators, control and sample is potentially biological infectious. The operator should comply with laboratory safe operation requirements and wear personal protective equipment (such as laboratory protective clothing, gloves, etc.).

10.2 Quality Control Material

Control will be used, which could be divided into level 1, level 2 and level 3.

Important

- Only specified quality control blood can be used.QC blood is designed to determine the state of the Analyzer.

10.3 Quality Control Methods

This analyzer use L-J QC, XBAR QC and X-M QC.

L-J QC: Single QC test data will be used daily to determine the performance of analyzer.

XBRA QC： Measure QC materials twice continuously every day, the mean value will be used to determine the performance of analyzer.

X-M QC: The mean value of the negative samples is calculated as the quality control point and is drawn on the quality control chart.

10.4 Preparation for QC

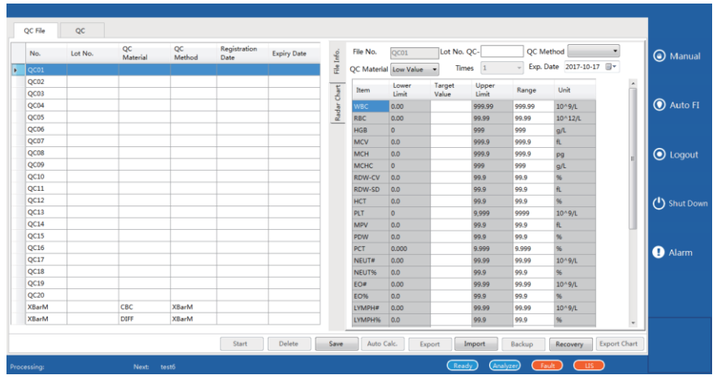
- Turn on the analyzer and wait for the mainframe to turn to measurement ready status.
- The QC file of new device will have no records.
- Create a QC file in the QC File interface.

10.5 QC File

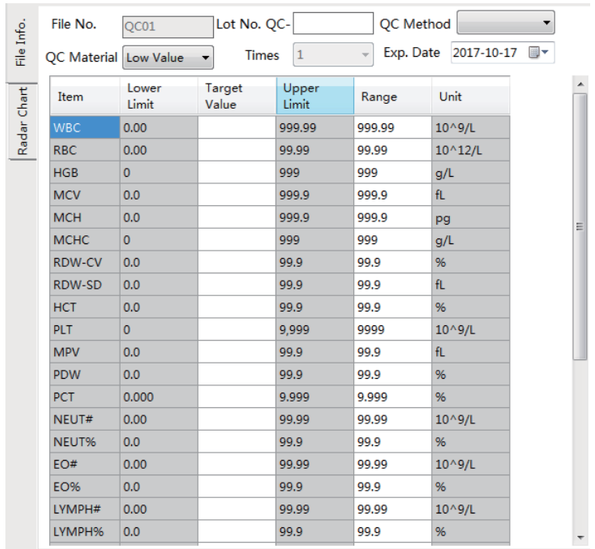
10.5.1 QC File Input

QC file must be registered before performing quality control tests. QC file input method is as follows:

1. Click "QC" button in main menu and then access to QC document interface, as Figure 10.5.1-1 shows.



2. Select a blank QC document, and then the corresponding document information will be indicated in right side of this interface, as Figure 10.5.1-2 shows.



3. Enter the lot number and expiry date on the control packaging, and target values, limits range and other information provided in the manual as Table 10.5.1-1 shows.

Table 10.5.1-1 Input specification

Name	Meaning and Instruction
Name	Analyzer series and model
Material	QC materials selected by user, including level1, level2,level3 and clinical samples.
Item	24 report parameters provided by this analyzer, except IG#, IG%, other#, other%.
Lower Limit	=Target Value - Limit Range
Target Vale	Manually set the input; obtained from the QC materials manual
Upper Limit	=Target Value + Limit Range
Lot No.	Control lot number printed on the package
File No.	QC file number
Unit	The corresponding units of items set in UMS
Exp. Date	On the control package, consistent with the expiry date of QC file
Save	Save the current input or modification and exit this interface

4. Click "Save" button to save and exit this interface after input is completed.

Important

- When the registered document is reentered again, it is modified.

10.5.2 QC File Modification

Click *"QC* Files" on the main menu to enter the QC file modification interface. Select the QC file number whose information needs modification in the file list and users can modify the lot number information of QC file in the lot input interface.

10.5.3 Move To Previous Data

Move the cursor in the file list to the previous file and select the QC file. Click "Previous" on the main menu or press T on the keyboard.

10.5.4 Move To Next Data

Move the cursor in the file list to the next file and select the QC file.
Click "Next" on the main menu or press 1 on the keyboard.

10.5.5 Delete

Select the QC file that needs deleting.

- Click to select a single file; Delete the selected file.
- When the file is selected, click "Delete" on the main menu and the confirmation dialog box pops out as Figure 10.5.5-1 shows.

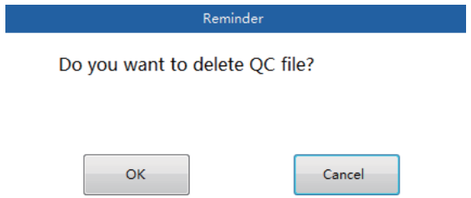


Figure 10.5.5-1

- Click "OK" to delete the selected QC file and close the dialog box.

Click "Cancel" to cancel QC file deletion and close the dialog box.

Important

- When the selected file does not have lot number information, The delete button disabled.

10.6 QC Chart

Use any of the following methods to enter QC chart interface:
In the QC file interface, double click the file number that needs graphic checking, and then click "QC chart" .
The data of the selected QC file will be displayed in fold line in chronological order on QC chart as Figure 10.6-1 shows.

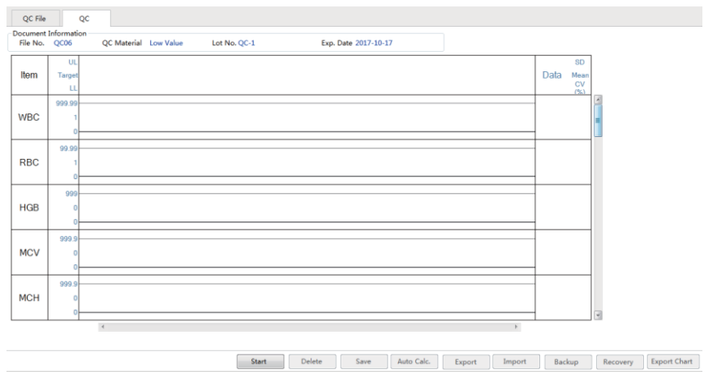


Figure 10.6-1

10.7 Perform QC analysis

QC analysis run in seal sampling mode. QC analysis consists of L-J QC, XBAR QC and X-M QC.

- Control is potentially biological infectious. The operator should comply with laboratory safe operation requirements and wear personal protective equipment (such as laboratory protective clothing, gloves, etc.

Perform QC analysis in manual mode as following steps.

1. Check whether the measurement ready status light on the mainframe is on. If it is, eliminate the fault first.

a) If the light is not on or is in other color, there may be some faults. Please click "Help" on the software interface to check and eliminate faults.

b) If the preparation light of instrument is on, the QC test can be performed.

2. When the instrument is ready, select a registered QC document and click "Start QC" on the toolbar to pop up the QC analysis interface.

10.7.1 L-J QC

L-J QC mode only run in manual sampling for once, shown as picture 10.7.1-

- After finishing measurement procedure, and then click "accept" button to finish this QC.

Workflow as follows:

1. Select one L-J QC document which has already registered, and then click "start QC" button enter into L-J QC interface.
2. After mixing quality controls totally, put the tube into manual sampling box. Press "manual sample" button to start QC analysis.
3. The results will be indicated in L-J QC interface after finishing analysis.

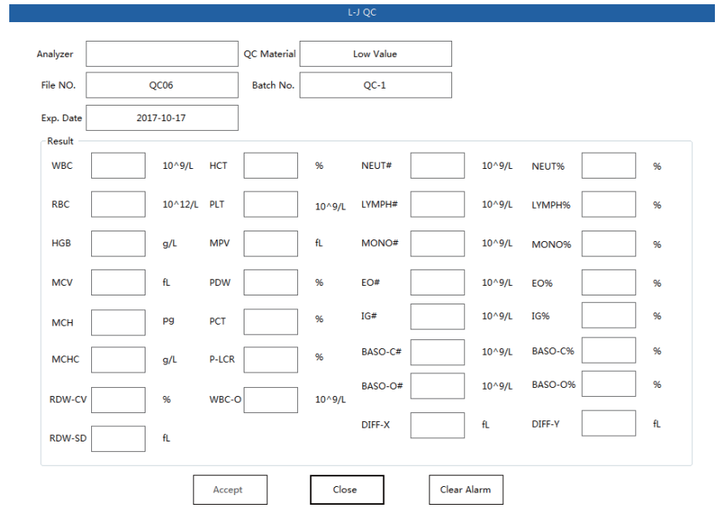


Figure 10.7.1 -1

10.7.2 XBAR QC

XBAR QC run in manual sampling mode for two times, shown as picture 10.7.2-1. After two times measurement, the instrument will get mean value, and then click ,accept " button to finish XBAR QC test.

Workflow as follows:

1. Select one XBAR QC document which has already registered, and then click "start QC button enter into XBAR QC interface.
2. After mixing quality controls totally, put the tube into manual sample box. Press "manual sample" button to start QC analysis for the first time.
3. The results will be indicated in XBAR QC interface after finishing analysis for the first time.
4. Press "manual sample" button to start QC analysis for the second time.
5. The results will be indicated in XBAR QC interface after finishing analysis for the second time.
6. The system will automatically calculate these two analysis results, and then get the QC test results.
7. Click "accept" button to finish QC.

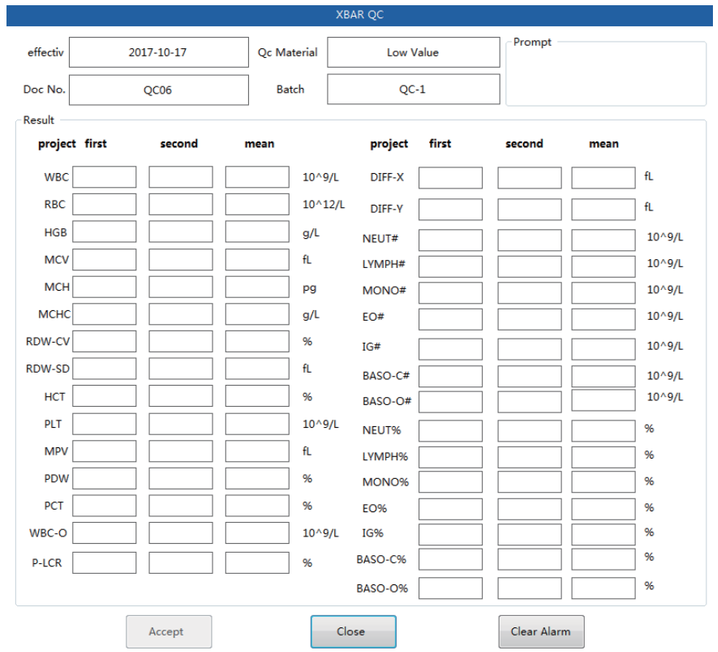


Figure 10.7.2-1

10.7.3 X-M QC

Use normal negative sample as QC materials, take average test results of certain number of samples as QC result and save it on the QC chart, you can choose whether or not to enable this QC method.

Workflow as follows:

1. Select "system" — "setting" — "QC", click "start X-M QC" button, set "X" For CBC sample number and "Y" for DIFF sample number,and then click "OK"
2. Fill in the target value and range information of X-M in QC.
3. Measure normal negative sample, every "X" CBC data will generate one X-M CBC information, and every "Y" DIFF data will generate one X-M DIFF information.
   The number of QC samples for CBC parameter can be different from DIFF parameter.

The CBC and DIFF parameters are placed in two QC folders separately, and QC materials can be regard as CBC and DIFF acquiescently.

X-M QC can only calculate the sample which meet the criteria after start the analyzer this time, if restart the analyzer, the previous sample cannot be listed in the calculation range, and one sample only can be calculated once

Important

■ X-M QC document do not have some function, such as back-up data, restore data, output target value, input target value, export document and import document.

10.7.4 QC Analysis Result Checking

1. After QC analysis, the analysis results will be displayed in the QC acceptance interface automatically.

There are several possible displays of the QC results:

- Results box after the item is empty " " , indicating that the item' s target value is not set.
- Results box after the item display " --------" ,indicating error in item testing
- Results box after the item displays red digit, indicating that the QC results exceed the set range upper and lower limits.

2. After the QC test, there are two buttons on the QC accepted interface for option.

Accept

Accept the QC test data, save it in the corresponding QC file database and draw corresponding QC data points in the QC chart.

Close

Cancel the QC test data, do not save it in the database of QC file.

3. View all QC data curves in QC chart interface. Monitor the performance of the device.

10.8 Abnormal results

This section is for reference only.

Parameter Alarm:

■ or displayed on the right side of parameter names means that the obtained analysis results exceed the set reference value range, but are

still within the measurement range of the analysis.

- displayed on the right side of parameter names means that the obtained analysis results is suspicious.
- displayed on the right side of parameter names means that the obtained analysis results exceed the linear measurement range.
- "----" displayed on the right side of parameter names means that the obtained analysis results are invalid.
- “,++++” displayed on the right side of parameter names means that the obtained analysis results exceed the display range.

Differential or Morphological Alarm:

WBC	**Criterion**
WBC Abnormal Scattergram	100%<(LYMPH%+MONO%+EO%+BASO%)
NRBC Abnormal Scattergram	Number of particles in Ghost ≥ 50
Neutropenia	Neut#<1*10^9/L or Neut%<0%
Neutrophilia	Neut#>11*10^9/L or Neut%>100%
Lymphopenia	Lymph#<0.8*10^9/L or Lymph%<0%
Lymphocytosis	Lymph#>4*10^9/L or Lymph%>100%
Monocytosis	Mono#>1*10^9/L or Mono%>100%
Eosinophilia	Eos#>0.7*10^9/L or Eos%>100%
Basophilia	Baso#>0.2*10^9/L or Baso%>100%
Leukocytopenia	WBC#<2.5*10^9/L
Leukocytosis	WBC#>18*10^9/L
Primitive cells	The point which SFL in LYMPH > mean+3sd is Blast in LYMPH, and the point which SFL in MONO > mean+sd is Blast in MONO.
Immature granulocytes	The particle which sfl > sfl+4sd in Neut should exceed a certain amount.
Move left	The percentage for the particle which sfl > sfl+4sd in Neut is exceed a certain value than total particle amount.

RBC	**Criterion**
Histogram of erythrocytes abnormal	The left and right boundary more 0.1 times than maximum value
Bimodal red blood cells	To find the trough in certain ranges
HGB abnormal	Rdwcv less than specific value and mcv less than specific value
Anisocytosis	RDW-SD>65fL or RDW-CV>20%
Microcytosis	MCV<70fL
Macrocytosis	MCV>110fL
Hypochromia	MCHC<29g/dL
Iron Deficiency?	HGB<100g/L
Erythrocytosis	RBC#>6.5*10^{12}/L

PLT	**Criterion**
Platelet histogram abnormal	The value for Rbcplt boundary more 0.4 times than the maximum value for pit data group, or the minimum value for pit data group more 0.1 times than the maximum value for ply data group, or pdw more than specific value
Thrombocytopenia	PLT<60*10^9/L
Thrombocytosis	PLT>600*10^9/L

Chapter 11 Customer Service

11.1 Overview

To ensure the accurate and efficient operation of the device, operators are required to conduct daily maintenance following this chapter instructions. The analyzer provides a variety of maintenance functions, which will facilitate operators maintenance.

Important	■ Improper maintenance might damage the analyzer. The analyzer must be maintained in accordance with the instructions. ■ If there are any problems unspecified in the instructions, please contact AGAPPE' s customer service for maintenance advice from professionals designated by AGAPPE. ■ The analyzer must be maintained by using components provided by AGAPPE. If you have any questions, please contact customer service department of our company.
☣	■ All the components surfaces of the analyzers are potentially biologically infectious, so safety precautions should be taken in the operating and maintenance process. ■ During operation, if blood or QC materials were spilled on the surface of the device, wipe with disinfection ethanol. During the cleaning process, safety precautions should be taken and laboratory safety requirements be followed strictly.

11.2 Maintenance

Click "System" — ''Maintenance" button, shown as follow figure 11.2-1.

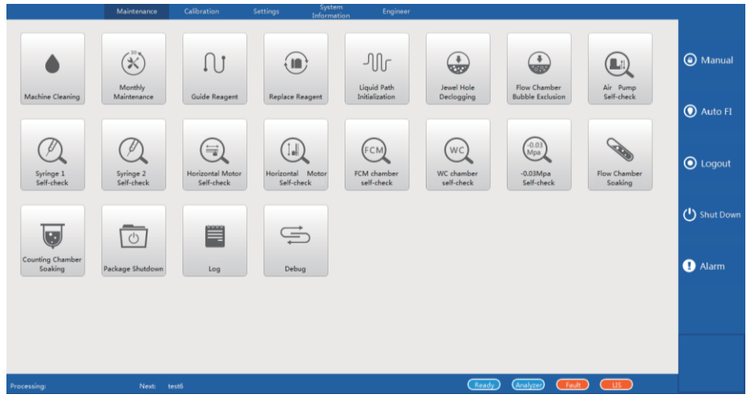


Figure 11.2-1

11.2.1 Whole Machine Cleaning

Click "Whole Machine Cleaning" , and the analyzer will automatically perform cleaning time sequence as Figure 11.2.1-1 shows. The dialog box will clear

automatically after cleaning.

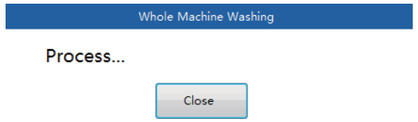


Figure 11.2.1-1

11.2.2 Jewel Hole Declogging

Click "Jewel Hole Declogging" to execute declogging time sequence. Analyzer backwashes jewl holes and burn them with a high voltage direct current on both sides to clear protein, dust and others clogged or adhered in the hole. Interface displays a progress bar of exclude plugging holes as Figure 11.2.2-1 shows.

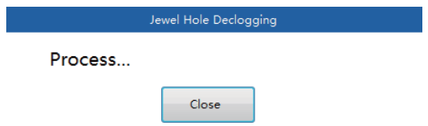


Figure 11.2.2-1

11.2.3 Flow Chamber Bubble Exclusion

Click "Flow Chamber Bubble Exclusion", and the analyzer will execute flow chamber to remove air bubbles time sequence, and displays a progress bar as Figure 11.2.3-1 shows.

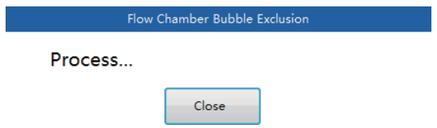


Figure 11.2.3-1

11.2.4 Liquid Path Initialization

Click Liquid Path Initialization" and the device will automatically perform the liquid path initializing, mechanical components initializing, and the device will be restored to its original state.

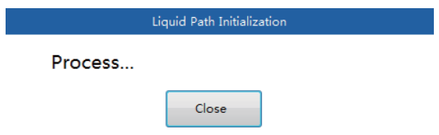


Figure 11.2.4-1

11.2.5 Sensor

Click "Sensors" to enter the sensor interface, as Figure 11.2.5-1 shows.

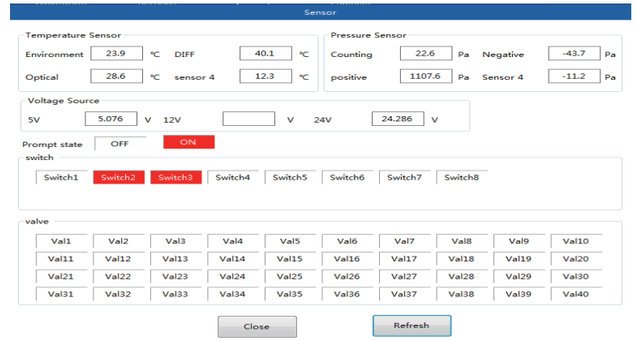


Figure 11.2.5-1

11.2.6 Monthly maintenance

Click ,Monthly Maintenance" and the Monthly Maintenance confirmation interface will pop out, as Figure 11.2.6-1 shows.

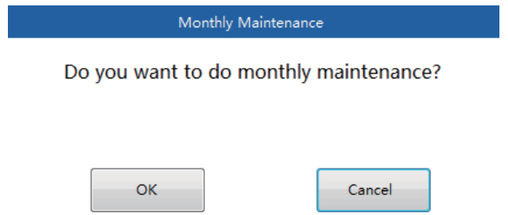


Figure 11.2.6-1

Click "Ok" to enter the Monthly Maintenance confirmation interface, as Figure 11.2.5-2 shows.

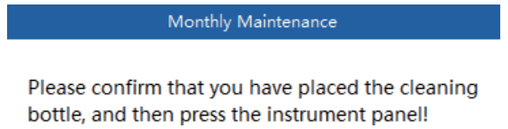


Figure 11.2.6-2

Press the button panel to pop up the interface, as shown in figure 11.2.6-3.

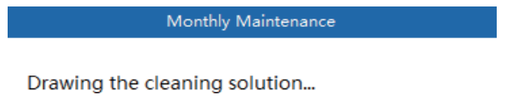


Figure 11.2.6-3

Important	■Monthly maintenance needs to draw cleaning fluid twice. Please follow the dialog box instructions.

11.2.7 Horizontal Motor Self-checking

Click Horizontal Motor Self-checking the automatically executes horizontal motor self-checking. If there are any problems, system will give motor abnormal alarm, as Figure 11.2.7-1 shows.

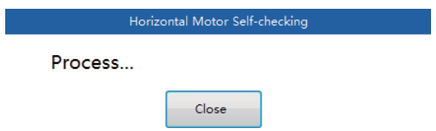


Figure 11.2.7-1

11.2.8 Vertical Motor Self-checking

Click "Vertical Motor Self-checking" the automatically executes vertical motor self-checking. If there are any problems, system will give motor abnormal alarm, as Figure 11.2.8-1 shows.

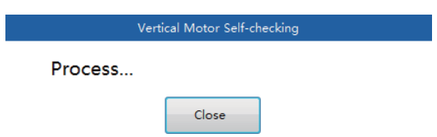


Figure 11.2.8-1

11.2.9 FCM Chamber Self-checking

Click "FCM Chamber Self-checking" and the analyzer automatically checks FCM chamber. If there are any problems, system will give FCM chamber abnormal alarm, as Figure 11.2.9-1 shows.

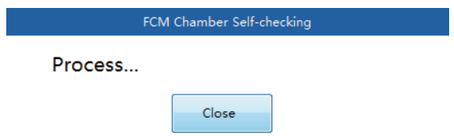


Figure 11.2.9-1

11.2.10 WC Chamber Self-checking

Click "WC Chamber Self-checking" and the analyzer automatically checks WC chamber. If there are any problems, system will give WC chamber abnormal alarm, as Figure 11.2.10-1 shows.

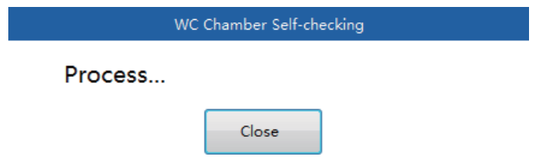


Figure 11.2.10-1

11.2.11 -0.03MPa Vacuum Self-checking

Click "-0.03MPa Self-checking" and the analyzer automatically executes -0.03MPa vacuum time sequence, as Figure 11.2.11-1 shows.

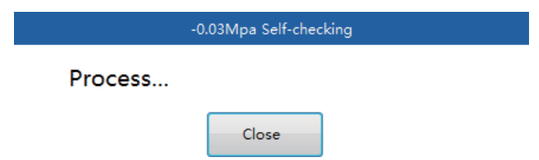


Figure 11.2.11-1

11.2.12 Reagent Replacement

Click " Reagent Replacement" to enter reagent replacement interface, as Figure 11.2.12-1 shows.

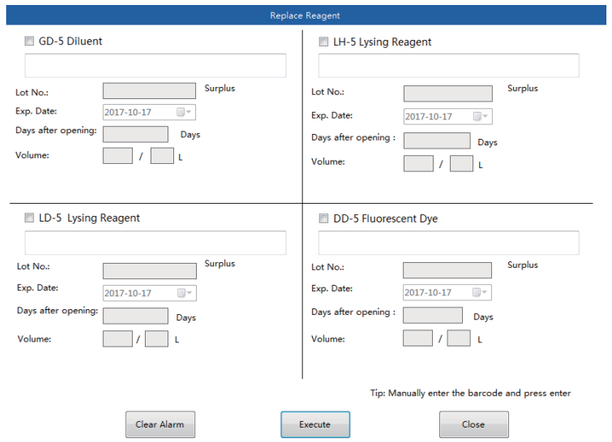


Figure 11.2.12-1

1) Check the box before the reagent that needs replacing.
2) In the "Manual Input" interface on the right, enter the new reagent's lot

number expiry date, expiry date after open and capacity(note the unit).

3) Click the "Execute" button, the instrument will lead the reagents automatically, and replace the capacity and the validity of the reagent in the database information.

4) Click "Close" after replacement and exit reagent replacement interface.

Important

■ When replacing a reagent, the capacity of the new reagent must be put in accurately and the units must be noted.

11.2.13 Reagent Guide

Click "Reagent Guide" to enter the reagent lead interface. Figure 11.2.13-1 shows.

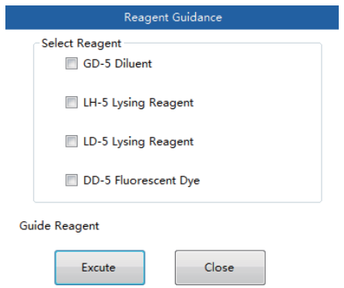


Figure 11.2.13-1

Select the box (user can select multiple reagents) before the reagent, click Execute and the device automatically guides the selected reagent(s).

Click "Shut down" after the guidance and exit reagent guide interface.

11.2.14 Air Pump Self-checking

Click "Air Pump Self-checking" and the analyzer executes air pump self-checking time sequence. If the air pump is abnormal, a prompt window will pop out. Figure 11.2.14-1 shows.

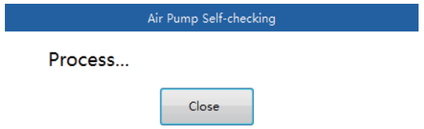


Figure 11.2.14-1

11.2.15 Counting Chamber Soaking

When there are wastes in the counting chamber, execute counting chamber soaking.

Click Counting Chamber Soaking" and the confirmation prompt interface pops out. Figure 11.2.15-1 shows.

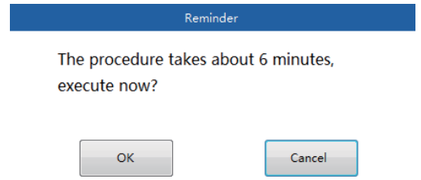


Figure 11.2.15-1

Click "ok" and the instruction dialogue window pops out.

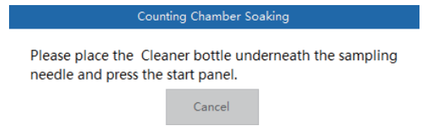


Figure 11.2.15-2

According to the instruction, place the cleaning fluid bottle under the aspiration pipette and press the key board, the device automatically draws cleaning fluid to soak, as Figure 11.2.15-3 shows.

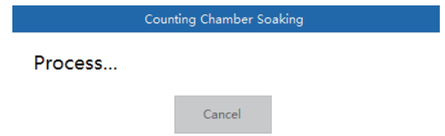


Figure 11.2.15-3

11.2.16 Flow Chamber Soaking

When the flow chamber is dirty, execute flow chamber soaking.

Click "Flow Chamber Soaking" and the confirmation prompt window pops out, as Figure 11.2.16-1 shows.

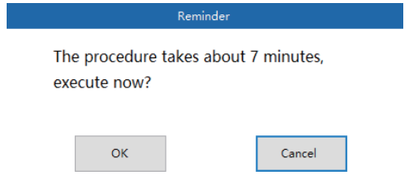


Figure 11.2.16-1

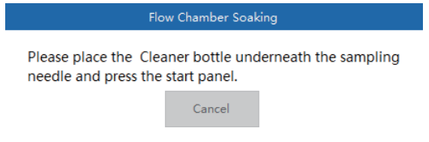


Figure 11.2.16-2

According to the dialog box, place the bottle in the bottom of the sampling pin and press the button plate, the instrument automatically absorbs the cleaning liquid to soak, and the pop-up interface as Figure 11.2.16-3 shows.

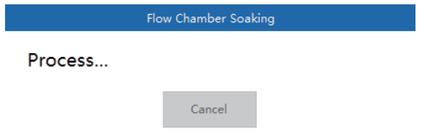


Figure 11.2.16-3

Chapter 12 Fault Handling

12.1 Overview

This chapter describes the possible faults that may occur during operation and provides appropriate solutions.

Important

- This manual is a non-maintenance manual, and only **important** provides solutions when the analyzer malfunctions and gives off alarm.

- Samples, calibrators, control, and waste liquid, etc. are potentially biological infectious. The operator should comply with laboratory safe operation requirements and wear personal protective equipment (such as laboratory protective clothing, gloves, etc.).

12.2 Fault Information and Processing

If abnormality is detected in using, the fault will be displayed in information display area of the status bar. Some of the fault information is accompanied with voice prompts.
When multiple faults appear simultaneously, user also can choose "Clear All" function, please follow as 12.2-1.

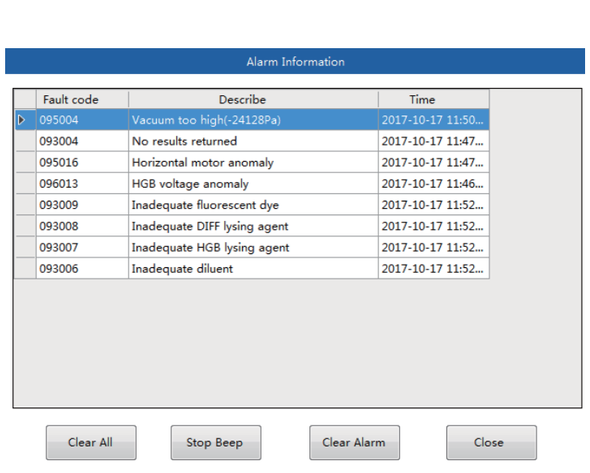


Figure 12.2-1

Fault No.	Fault Description	Executive action	Handling Measures
1	Counting pressure is unstable	In the process of automatic injection, the malfunction occurs and the instrument is suspended	1) Double-click the fault information, detailed fault processing interface will open, click "Eliminate" . If the fault cannot be eliminated, please 2) contact customer services department of our company or local dealers.
2	Startup DIFF chamber temperature does not reach the standard	When a fault occurs, the instrument is prompted to operate	1) Reboot device. 2) If the fault cannot be eliminated, please contact customer services department of our company or local dealers.
3	Startup optical system temperature does not reach the standard	When a fault occurs, the instrument is prompted to operate	1) Reboot device. 2) If the fault cannot be eliminated, please contact customer services department of our company or local dealers.
4	Device operation timed out	When a fault occurs, the instrument is prompted to operate	1) Double-click the fault information, detailed fault processing interface will open, click " Eliminate" . 1) If the fault cannot be eliminated, please contact customer services department of our company or local dealers.
5	HGB background voltage is abnormal	In the process of automatic injection, the malfunction occurs and the instrument is suspended	1) Double-click the fault information, detailed fault processing interface will open, click "Eliminate" . 1) If the fault cannot be eliminated, please contact AGAPPE customer services department of our company or local dealers.
6	Diluent insufficient	In the process of automatic injection, the malfunction occurs and the instrument is suspended	1) Double-click the fault information, detailed fault processing interface will open, click "Eliminate" to enter reagent replacement interface. Replace the diluent.
7	LH-5 lysing reagent insufficient	In the process of automatic injection, the malfunction occurs and the instrument is suspended	1) Double-click the fault information, detailed fault processing interface will open, click "Eliminate" to enter reagent replacement interface. Replace the LH-5 lysing reagent, the fault eliminated automatically.

8	LD-5 lysing reagent insufficient	In the process of automatic injection, the malfunction occurs and the instrument is suspended	1) Double-click the fault information, detailed fault processing interface will open, click “Eliminate” to enter reagent replacement interface. Replace the LD-5 lysing reagent, the fault eliminated automatically
9	Fluorescent dye insufficient	In the process of automatic injection, the malfunction occurs and the instrument is suspended	1) Double-click the fault information, detailed fault processing interface will open, click "Eliminate to enter reagent replacement interface. Replace the fluorescent dye.
10	Diluent expired	In the process of automatic injection, the malfunction occurs and the instrument is suspended	1) Double-click the fault information, detailed fault processing interface will open. Click "Eliminate” to enter reagent replacement interface. Replace the diluents.
11	LH-5 lysing reagent expired	In the process of automatic injection, the malfunction occurs and the instrument is suspended	1) Double-click the fault information, detailed fault processing interface will open. Click “Eliminate” to enter reagent replacement interface. Replace the new LH-5 lysing reagent.
12	LD-5 lysing reagent expired	In the process of automatic injection, the malfunction occurs and the instrument is suspended	1) Double-click the fault information, detailed fault processing interface will open. Click “Eliminate" to enter reagent replacement interface. Replace the new LD-5 lysing reagent.
13	Fluorescent dye expired	In the process of automatic injection, the malfunction occurs and the instrument is suspended	1) Double-click the fault information, detailed fault processing interface will open. Click "Eliminate” to enter reagent replacement interface. Replace the fluorescent dye.
14	Upper and lower machine edition mismatch	No processing	1) Please contact customer services department of our company or local dealers.
15	Blank check does not reach the standard	Restart	1) Conduct blank check again. 1) Conduct machine wash. 2) Conduct monthly maintenance. 3) If the fault cannot be eliminated, please contact customer services of our company or local dealers.
16	Channel data collection error	No processing	1) Double-click the fault information, detailed fault processing interface will open, click "Eliminate” . 1) If the fault cannot be eliminated, please contact customer services of our company or local dealers.
17	Data storage failed	When a fault occurs, the instrument is prompted to operate	1) Double-click the fault information, detailed fault processing interface will open, click "Eliminate” . 2) Reboot upper and lower machine. 3) If the fault cannot be eliminated, please contact customer services of our company or local dealers.
18	Thread start failed	When a fault occurs, the instrument is prompted to operate	1) Reboot upper and lower machine. 2) If the fault cannot be eliminated, please contact customer services of our company or local dealers.
19	Serial port operation failed	When a fault occurs, the instrument is prompted to operate	1) Reboot upper and lower machine. 1) If the fault cannot be eliminated, please contact customer services of our company or local dealers.
20	System file opening failed	When a fault occurs, the instrument is prompted to operate	1) Reboot upper and lower machine. If 2) the fault cannot be eliminated, please contact customer services our company or local dealers.
21	Version reading failed	When a fault occurs, the instrument is prompted to operate	1) Reboot upper and lower machine. 2) If the fault cannot be eliminated, please contact customer services of our company or local dealers.
22	Memory error	When a fault occurs, the instrument is prompted to operate	1) Reboot upper and lower machine. 2) If the fault cannot be eliminated, please contact customer services of our company or local dealers.
23	Time sequence download failed	When a fault occurs, the instrument is prompted to operate	1) Reboot upper and lower machine. 2) If the fault cannot be eliminated, please contact customer services of our company or local dealers.

24	Unknown command	Downtime	1) Reboot upper and lower machine. If the 2) fault cannot be eliminated, please contact customer services of our company or local dealers.
25	Communication with the device off! Reboot the device	Downtime	1) Reboot upper and lower machine. 2) If the fault cannot be eliminated, please contact customer services of our company or local dealers.
26	Time sequence timeout, please clear the fault in fault elimination interface	When a fault occurs, the instrument is prompted to operate	1) Reboot upper and lower machine. 2) If the fault cannot be eliminated, please contact customer services of our company or local dealers.
27	Serial port communication error	Downtime	1) Reboot upper and lower machine. If the 2) fault cannot be eliminated, please contact customer services department of our company or local dealers.
28	TCP communication error	Downtime	1) Reboot upper and lower machine. If the 2) fault cannot be eliminated, please contact customer services department of our company or local dealers.
29	Function time sequence not found	Downtime	1) Reboot upper and lower machine. If the 2) fault cannot be eliminated, please contact AGAPPE customer services or local dealers.
30	Sub time sequence not found	Downtime	1) Reboot upper and lower machine. If the 2) fault cannot be eliminated, please contact AGAPPE customer services or local dealers.
31	Function instruction execution time-out	Downtime	1) Please contact AGAPPE customer services or local dealers.
32	Pressure is too high	Downtime	1) Double-click the fault information, detailed fault processing interface will open, click "Eliminate" . 2) Perform vacuum pump self- checking procedures. 3) If the fault cannot be eliminated, please contact AGAPPE customer services or local dealers.
33	Pressure is too low	Downtime	1) Double-click the fault information, detailed fault processing interface will open, click "Eliminate" . 2) Perform vacuum pump self-checking procedures. 3) If the fault cannot be eliminated, please contact AGAPPE customer services or local dealers.
34	High vacuum	Downtime	1) Double-click the fault information, detailed fault processing interface will open, click "Eliminate". 2) Perform vacuum pump self-checking procedures. 3) If the fault cannot be eliminated, please contact AGAPPE customer services or local dealers.
35	Low vacuum	Downtime	1) Double-click the fault information, detailed fault processing interface will open, click "Eliminate". 2) Perform vacuum pump self-checking procedures. 3) If the fault cannot be eliminated, please contact AGAPPE customer services or local dealers.
36	DIFF chamber temperature is too high	Report on startup	1) Please contact AGAPPE customer services or local dealers.
37	DIFF chamber temperature is too low	Report on startup	1) Please contact AGAPPE customer services or local dealers.
38	DIFF chamber temperature sensor is broken	When the fault occurs in the automatic measurement, the instrument is able to measure the sample that is currently being measured and will stop operation	1) Please contact AGAPPE customer services or local dealers.
39	Optical system temperature is too high	Report on startup	1) Please contact AGAPPE customer services or local dealers.
40	Optical system temperature is too low	Report on startup	1) Please contact AGAPPE customer services or local dealers.

41	Optical system temperature sensor is broken	Delay shutdown	1) Please contact AGAPPE customer services or local dealers.
42	Environment temperature is too high	Report on startup	1) Observe whether the indoor temperature is within the allowable range of the device. 2) If not, please turn on the A/C to make indoor temperature meet the requirements.
43	Environment temperature is too low	Report on startup	1) Observe whether the indoor temperature is within the allowable range of the device. 2) If not, please turn on the A/C or heating to make indoor temperature meet the requirements.
44	Environment temperature sensor is broken	Delay shutdown	1) Please contact AGAPPE customer services or local dealers.
45	Horizontal motor is abnormal	The instrument will stop immediately when there is a failure in the automatic measurement	1) Double-click the fault information, detailed fault processing interface will open, click "Eliminate". If the fault cannot be eliminated, 2) please contact AGAPPE customer services or local dealers.
46	Vertical motor is abnormal	The instrument will stop immediately when there is a failure in the automatic measurement	1) Double-click the fault information, detailed fault processing interface will open, click "Eliminate". If the fault cannot be eliminated, 2) please contact AGAPPE customer services or local dealers.
47	Syringe 1 is abnormal	The instrument will stop immediately when there is a failure in the automatic measurement	1) Double-click the fault information, detailed fault processing interface will open, click "Eliminate". 2) If the fault cannot be eliminated, please contact AGAPPE customer services or local dealers.
48	Syringe 2 is abnormal	The instrument will stop immediately when there is a failure in the automatic measurement	1) Double-click the fault information, detailed fault processing interface will open, click "Eliminate". If the fault cannot be eliminated, please 2) contact AGAPPE customer services or local dealers.
49	Communication abnormal	The instrument will stop immediately when there is a failure in the automatic measurement	1) Reboot upper and lower machine. If the 2) fault cannot be eliminated, please contact AGAPPE customer services or local dealers.
50	Waste tank is not empty	The instrument will stop immediately when there is a failure in the automatic measurement	1) Double-click the fault information, detailed fault processing interface will open, click "Eliminate". 2) If the fault cannot be eliminated, please contact AGAPPE customer services or local dealers.
51	Waste bucket full	In the process of automatic injection, the malfunction occurs and the instrument is suspended.	1) Empty the waste bucket. 2) Double-click the fault information, detailed fault processing interface will open, click "Eliminate". 3) If the fault cannot be eliminated, please contact AGAPPE customer services or local dealers.
52	FCM chamber is not full	In the process of automatic injection, the malfunction occurs and the instrument is suspended.	1) Check whether there are enough diluents. Double-click the fault information, 2) detailed fault processing interface will open, click "Eliminate". 3) If the fault cannot be eliminated, please contact AGAPPE customer services or local dealers.
53	Left door open	Standby time prompt, run time stop	1) Close the left door. 2) Double-click the fault information, detailed fault processing interface will open, click "Eliminate". 3) If the fault cannot be eliminated, please contact AGAPPE customer services or local dealers.

54	Right door open	Standby time prompt, run time stop	1) Close the right door. 2) Double-click the fault information, detailed fault processing interface will open, click "Eliminate". 3) If the fault cannot be eliminated, please contact AGAPPE customer services or local dealers.
55	Optical shielding box open	Standby time prompt, run time stop	1) Close the optical shielding box. Double- 2) click the fault information, detailed fault processing interface will open, click "Eliminate". 3) If the fault cannot be eliminated, please contact AGAPPE customer services or local dealers.
56	Laser current is abnormal	In the process of automatic injection, the malfunction occurs and the instrument is suspended	1) Please contact AGAPPE customer services or local dealers.
57	+12V voltage is abnormal	In the process of automatic injection, the malfunction occurs and the instrument is suspended	1) Please contact AGAPPE customer services or local dealers.
58	-12V voltage is abnormal	In the process of automatic injection, the malfunction occurs and the instrument is suspended	1) Please contact AGAPPE customer services or local dealers.
59	+5V voltage is abnormal	In the process of automatic injection, the malfunction occurs and the instrument is suspended	1) Please contact AGAPPE customer services or local dealers.
60	RBC jewel hole clogging	In the process of automatic injection, the malfunction occurs and the instrument is suspended	1) Double-click the fault information, detailed fault processing interface will open, click "Eliminate". 2) Execute counting chamber soaking. 3) If the fault cannot be eliminated, please contact AGAPPE customer services or local dealers.
61	WBC jewel hole clogging	In the process of automatic injection, the malfunction occurs and the instrument is suspended	1) Double-click the fault information, detailed fault processing interface will open, click "Eliminate". 2) Execute counting chamber soaking. 3) If the fault cannot be eliminated, please contact AGAPPE customer services or local dealers.
62	HGB voltage is abnormal	In the process of automatic injection, the malfunction occurs and the instrument is suspended	1) Double-click the fault information, detailed fault processing interface will open, click "Eliminate". 2) Execute counting chamber soaking. 3) If the fault cannot be eliminated, please contact AGAPPE customer services or local dealers.

Appendix Specification

1.Vacuum vessel type

	Diameter (Tube body)	Length	Remarks
Standard	12mm	75mm	/
Allowed diameter		/	/

2.Parameter Description

Abbreviation	Name	Default Units
WBC	White Blood Cell Count	10^9/L
RBC	Red Blood Cell Count	10^{12}/L
HGB	Hemoglobin	g/L
MCV	Mean Corpuscular Volume	fL
MCH	Mean Corpuscular Hemoglobin	pg
MCHC	Mean Corpuscular Hemoglobin Concentration	g/L
RDW-CV	RBC Distribution Width-Coefficient of Variation	%
RDW-SD	RBC Distribution Width-Standard Deviation	fL
HCT	Hematocrit	%
PLT	Platelet Count	10^9/L
P-LCR	Large platelet ratio	%
MPV	Mean Platelet Volume	fL
PDW	Platelet Distribution Width	%
PCT	Platelet Hematocrit	%
BASO#	Basophilic Granulocyte	10^9/L
BASO%	Basophilic Granulocyte Percentage	%
NEUT#	Neutrophile Granulocyte	10^9/L
NEUT#	Neutrophile Granulocyte	%
EO#	Eosinophilic Granulocyte	10^9/L

EO%	Eosinophilic Granulocyte Percentage	%
LYMPH#	Lymphocyte	10^9/L
LYMPH%	Lymphocyte Percentage	%
MONO#	Monocyte	10^9/L
MONO%	Monocyte Percentage	%
IG #	Immature Granulocyte (research)	10^9/L
IG%	Immature Granulocyte Percentage (research)	%
OTHER#	Other Cell (research)	10^9/L
OTHER%	Other Cell Percentage (research)	%
WBC Histogram	White Blood Cell Histogram	None
RBC Histogram	Red Blood Cell Histogram	None
PLT Histogram	Platelet Histogram	None
DIFF Scattergram	Scattergram	None

3.Sample Characteristics

- Sample required for each analysis (CBC+Diff)
  Open sampling-whole blood 20u L
  Open sampling-pre-dilution 91.5u L
- Dilution Ratio

	WBC/HGB	RBC/PLT	WBC Differentiation
Whole blood	1:349	1:16720	1:140
Capillary blood	1:627	1:32425	1:254

- Measurement Time
  Manual sampling: w 60 seconds / test

4.Performance Indicators

- Background Range

Parameters	Blank Counting
WBC	≤ 0.5 X 109/L
RBC	≤ 0.05X10^{12}/L
HGB	≤ 2g/L
PLT	≤ 10X10^9/L

- Linearity Range

Parameters	Linear Measurement Range	Linear Tolerance	r
WBC	1.0X10^9/L ~ 10.0X10^9/L	Less than ± 0.5 x 10^9/L	≥ 0.990
	10.1 X10^9/L ~ 99.9X10^9/L	Less than ± 5.0%	
RBC	0.30X10^{12}/L ~ 1.00X10^{12}/L	Less than ± 0.05 *x* 10^{12}/L	≥ 0.990
	1.01 X10^{12}/L ~ 7.00X10^{12}/L	Less than ± 5.0%	
HGB	20g/L ~ 70g/L	Less than ± 2g/L	≥ 0.990
	71g/L - 240g/L	Less than ± 3%	
PLT	20x10^9/L ~ 100x10^9/L	Less than ± 10 x 10^9/L	≥ 0.990
	101 X10^9/L ~ 999X10^9/L	Less than ± 10.0%	

- **Repeatability**

Parameters	**Repeatability**	**Conditions**
WBC	≤ 4.0%	WBC (4.0-10.0 X10^9/L)
RBC	≤ 2.0%	RBC (3.50-5.50 x10^{12}/L)
HGB	≤ 2.0%	HGB (110-160 g/L)
HCT/MCV	≤ 3.0% ≤ 3.0%	MCV (35% ∽50%, 80fL ∽100fL)
PLT	≤ 8.0%	PLT (100-300 x10^9/L)

- **Accuracy**

Parameter	**Allowed bias range**
WBC	Less than ±5.0%
RBC	Less than ±2.0%
HGB	Less than ± 2.0%
PLT	Less than ± 8.0%
HCT/MCV	Less than ± 2.0%

- **Carryover Rate**

Parameter	**Carryover Rate**
WBC	≤ 3.5%
RBC	≤ 2.0%
HGB	≤ 2.0%
PLT	≤ 5.0%

5.Boundary Dimension and Weight

	Analyzer Mainframe
Width	390mm
Deep	480mm
Height	530mm
Weight	31kg

6.Accessories list

Item No.	Quantity	Accessories Name
1	1	Waste Fluid Bucket Soft Bag
2	1	Operator's Manual
3	1	Ground Wire
4	1	Software Installation Thumb Drive
5	1	External Waste Fluid Bucket Tube
6	1	External Diluent Tube
7	1	External LD-5 Lysing Reagent Tube
8	1	External LH-5 Lysing Reagent Tube
9	1	British standard power cable
10	1	American standard power cable
11	1	European standard power cable
12	1	Cable
13	1	Cleaning Fluid (optional)
14	1	Handheld barcode scanner (optional)

EN ISO 13485:2016 CE

EU REP

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